

Equidistribution of Primitive Sextic Torus Packets

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Abstract

We prove that the complete ideal-class packets of maximal orders in totally real sextic fields with no proper intermediate fields become equidistributed, with no escape of mass, in the space of unimodular lattices as their field discriminants tend to infinity. The limit is Haar probability measure. Each packet includes all coordinate-sign translates and is weighted by diagonal-orbit volume.

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1 Introduction

The real embeddings of a totally real number field turn its ideals into lattices. Multiplication by positive diagonal matrices changes the relative sizes of the embedding coordinates while preserving covolume. The units make each resulting diagonal orbit compact, and the ideal classes assemble these orbits into a finite packet. The distribution of such packets connects the arithmetic of number fields with the dynamics of higher-rank diagonal groups. We prove equidistribution for totally real sextic fields having no proper intermediate field.

1.1 Packets and the main theorem

Let

$$X_6 = \mathrm{SL}_6(\mathbb{Z}) \backslash \mathrm{SL}_6(\mathbb{R}), \quad A_6 = \{\mathrm{diag}(e^{u_1}, \dots, e^{u_6}) : \sum_j u_j = 0\}.$$

We regard X_6 as the space of covolume-one lattices in $\mathbb{R}^6$, with matrices acting on row vectors on the right. Let m_6 denote its $\mathrm{SL}_6(\mathbb{R})$ -invariant probability. We normalize Haar measure on A_6 as $du_1 \cdots du_5$, with $u_6 = -\sum_{j=1}^5 u_j$.

A degree-six number field K is *primitive* if there is no field F with $\mathbb{Q} \subsetneq F \subsetneq K$. Suppose K is totally real, choose an ordering $\sigma = (\sigma_1, \dots, \sigma_6)$ of its real embeddings, and write $\mathcal{O}_K$ for its ring of integers. For a nonzero fractional $\mathcal{O}_K$ -ideal I , set

$$\sigma(I) = \{(\sigma_1(d), \dots, \sigma_6(d)) : d \in I\}, \quad \Lambda_{I,\sigma} = \mathrm{covol}(\sigma(I))^{-1/6} \sigma(I).$$

Choose one representative of each ordinary ideal class. Let $\mathcal{P}_{K,\sigma}$ be the set of distinct compact orbits

$$\Lambda_{I,\sigma} w A_6, \quad w = \mathrm{diag}(\epsilon_1, \dots, \epsilon_6), \quad \epsilon_j \in \{1, -1\}.$$

Here every sign matrix acts on the underlying lattice, including those of determinant -1 . Dirichlet's unit theorem gives compactness of the orbits. Including all signs makes the packet independent of the chosen ordinary ideal-class representatives.

For $\mathcal{O} \in \mathcal{P}_{K,\sigma}$, let $m_{\mathcal{O}}$ be its invariant probability and $\mathrm{vol}_A(\mathcal{O})$ its volume for the specified Haar measure. The packet probability is

$$\mu_{K,\sigma} = \frac{\sum_{\mathcal{O} \in \mathcal{P}_{K,\sigma}} \mathrm{vol}_A(\mathcal{O}) m_{\mathcal{O}}}{\sum_{\mathcal{O} \in \mathcal{P}_{K,\sigma}} \mathrm{vol}_A(\mathcal{O})}. \quad (1.1)$$

Theorem 1.1. *Let K_i be totally real primitive fields of degree six, with $|\mathrm{Disc}(K_i)| \rightarrow \infty$, and choose any ordering σ_i of the embeddings of each field. Then*

$$\int_{X_6} f \, d\mu_{K_i, \sigma_i} \longrightarrow \int_{X_6} f \, dm_6 \quad (f \in C_c(X_6)).$$

There is no escape of mass: for every $\epsilon > 0$ there is a compact $C \subset X_6$ such that $\mu_{K_i, \sigma_i}(C) \geq 1 - \epsilon$ for all sufficiently large i .

The theorem concerns each complete packet attached to a field. It allows the fields and their embedding orderings to vary freely under the stated hypotheses. The arithmetic role of primitivity is to make every nonrational integer of K_i generate the full field; this will yield uniform separation in the dual lattice used to count thin tubes.

1.2 History and the composite-degree obstruction

For real quadratic fields, the corresponding packets project to packets of closed geodesics on the modular surface. Linnik's ergodic method, including Skubenko's treatment of the one-sheeted hyperboloid, gave equidistribution under an auxiliary splitting condition [12, Chapter VI]; see also the historical statements in [10, Theorems 1.1–1.2]. Duke removed this condition for fundamental discriminants by analytic methods [3, Theorem 1]. Einsiedler, Lindenstrauss, Michel, and Venkatesh later gave a dynamical proof formulated for all positive nonsquare order discriminants [10, Theorem 2.3]. In higher degree the compact diagonal orbits have dimension greater than one, and the dynamics of the full diagonal group provides additional rigidity.

Einsiedler, Katok, and Lindenstrauss proved that, on $\mathrm{SL}_d(\mathbb{Z}) \backslash \mathrm{SL}_d(\mathbb{R})$ for $d \geq 3$, an ergodic probability invariant under the full positive diagonal group and having positive entropy for a diagonal flow is algebraic [7, Theorem 1.3]. Einsiedler, Lindenstrauss, Michel, and Venkatesh developed the connection between arithmetic separation of periodic torus orbits and entropy of their limiting measures [8]. Their subsequent cubic equidistribution theorem combined measure classification, subconvexity, and local harmonic analysis [9]. These works distinguish two tasks that remain central here: establishing nonescape and entropy, and identifying the algebraic components of a limit.

The second task is particularly important in composite degree [9, Section 1.6.3]. In prime degree, the positive-entropy classification gives Haar measure directly [7, Corollary 1.4]. In degree six there are proper block subgroups containing the full diagonal group whose closed orbits can carry finite invariant measures. Primitivity of the fields in the sequence does not, by itself, rule out concentration near such orbits. The proof must retain enough information from the sequence to exclude these limiting components.

Subsequent work has sharpened the entropy estimates in higher degree. Khayutin obtained bounds for maximal-order packets whose Galois closures act two-transitively on the embeddings [11, Theorem 1.1]. Wieser and Yang obtained entropy bounds for quartic packets of maximal type with quadratic subfields, under a fixed split-place condition and additional discriminant-growth hypotheses [19]. These results clarify the arithmetic information available in composite degree; identifying every component of a limit remains a separate step in the argument here.

The arithmetic preparation adapts the ordinary-vector method used for prime-degree packets [13, Sections 2–6] and primitive quartic packets [14, Section 3]. We give the needed estimates in full. They use Stark’s effective zero-free information [18] and the short-interval bounds of Shiu and Pollack [15, 17]. The quartic work treats its additional obstruction using exterior-square arithmetic. Here an ordinary vector and a dual vector produce a codifferent element, and primitivity gives the separation needed to estimate its partial traces.

1.3 The intermediate-resolution argument

The main additional argument concerns conditional measures that retain information at growing, but separated, resolutions. We describe its purpose before introducing the construction.

Let $\mu_i = \mu_{K_i, \sigma_i}$. Ordinary-vector estimates prove tightness and give every probability limit μ an $O(r^6)$ bound for the mass of radius- r balls, uniformly when their centers range over a compact set. Since A_6 has dimension five, this bound forces positive entropy in almost every A_6 -ergodic component. Theorem 3.2 then makes those components homogeneous. Every proper component is supported on points fixed by a nonidentity diagonal matrix with algebraic entries, so only a countable family of fixed-point sets must be excluded.

The arithmetic estimate used for this exclusion concerns two-sided tubes, with O a sufficiently small identity neighborhood,

$$x(a(tb)Oa(-tb) \cap a(-tb)Oa(tb)), \quad a(b) = \mathrm{diag}(e^{b_1}, \dots, e^{b_6}), \quad \sum_j b_j = 0.$$

If the coordinates of b are separated across a prescribed cut of the indices, ordinary and dual vectors bound the packet mass of such a tube by $e^{-\eta t}$ for a fixed $\eta > 0$. The estimate holds throughout a growing interval $[T_{-,i}, T_{+,i}]$ with $T_{+,i}/T_{-,i} \rightarrow \infty$. The breadth of this interval is used throughout the proof.

For $j \neq k$, let $U_{jk} = \{1 + sE_{jk} : s \in \mathbb{R}\}$. Exact conditional measures of a finite packet along these root groups are supported at a single point. They therefore cannot express the spread detected by the arithmetic tube bound. Instead, in a local box (q, h) for a unipotent group, we condition the

plaque coordinate h on a small cell of the transverse coordinate q . We choose logarithmic cell depths inside the growing interval so that the conditional probabilities change negligibly across a much wider internal range. In their joint limits with the sampled lattice, the plaque point still has the recorded probability as its conditional law, given the transverse coordinate and that one probability. This sampling identity will detect atoms on sets meeting each root plaque in only countably many points. The limiting probabilities also retain their behavior under suitable long diagonal motions.

Product structure for genuine leafwise measures and the use of noncommuting root foliations are basic ingredients of the higher-rank entropy method of Einsiedler and Katok [4, Proposition 8.3] [5, Theorems 8.4–8.5]; see also [6, Corollary 8.8 and Theorem 9.8]. The point here is to prove the corresponding structure for the conditional probabilities formed before taking the limit. If a diagonal element fixes one root group U and contracts a complementary unipotent subgroup W generated by root groups and normal in WU , the limiting measure on WU is the product of the limiting measures on W and U . Contraction identifies the U factor. A separate conditional-information argument shows that adding a fine U -cell label barely changes the law of any fixed further subdivision of a typical W tile. Expanding that tile turns this comparison into one for bounded W tests and identifies the W factor. Both arguments are needed, and neither follows by simply disintegrating the weak limit.

Call $j \rightarrow k$ active when the resulting U_{jk} measure has support beyond the identity. The product rule makes activity transitive under composition of roots. It also permits an entropy argument: if crossing roots are inactive along a positive proportion of long trajectories, then their names can be encoded using only the slowly expanding directions inside the two sides of a cut. This contradicts the tube bound. Long motions in the diagonal wall $b_j = b_k$ fixing U_{jk} leave its activity unchanged. Combining the resulting one-dimensional functions of the differences $b_k - b_j$ upgrades activity along trajectories to full activity at almost every limiting state. A proper homogeneous fixed-point piece forces an inactive root and is therefore excluded.

The construction separates the arithmetic estimate from this last argument. Its potentially reusable ingredients are the comparison of conditional probabilities across a broad range of transverse depths, their product rule, and the passage from entropy bounds for long segments starting in any fixed positive proportion of a packet to full root activity.

Section 2 proves the arithmetic bounds, and Section 3 identifies the homogeneous alternatives. Sections 4 and 5 construct and factor the intermediate conditional measures. Section 6 records their root-group consequences. Sections 7 and 8 prove the long-segment and full-activity assertions, respectively, and Section 9 concludes the proof.

1.4 Conventions

For the remainder of the paper write $n = 6$, $G = \mathrm{SL}_n(\mathbb{R})$, $\Gamma = \mathrm{SL}_n(\mathbb{Z})$, $X = \Gamma \backslash G$, and $A = A_n$. Let $L = \{b \in \mathbb{R}^n : \sum_j b_j = 0\}$, identified with the traceless diagonal matrices, and $a(b) = \mathrm{diag}(e^{b_1}, \dots, e^{b_n})$. We use the right action throughout. Thus motion by $a(b)$ changes a relative displacement h to $a(-b)ha(b)$. Matrix neighborhoods of the identity and smooth metrics are interchangeable on fixed compact sets. When a metric on all of X is needed, we use the quotient of a left-invariant Riemannian metric on G . The notation $B(r)$ denotes a sufficiently small identity neighborhood of radius r . Implicit constants may depend on fixed compact sets, charts, and n , but not on the varying field. All passages to subsequences preserve the probability limit under consideration.

2 Arithmetic estimates for the packets

This section establishes the three estimates on which the dynamical argument rests: uniform control of the cusp, a dimension bound for probability limits, and exponential decay on two-sided diagonal tubes. The first two follow by counting ordinary lattice vectors. For the third, we count an ordinary vector together with a dual vector; their coordinatewise product lies in the codifferent and can be counted in a thin slice.

For the volume, unfolding, and ordinary-vector estimates, we adapt the method of [13, Sections 2–6] and [14, Section 3]. We reproduce the needed arguments in the packet and Haar normalizations of this paper.

Throughout this section, K is a totally real field of degree six with no proper intermediate field, and σ is an ordering of its real embeddings. Write

$$D = |\text{Disc}(K)|, \quad Q = D^{1/2}, \quad \kappa = \text{Res}_{s=1} \zeta_K(s), \quad d_K(l) = \#\{\mathfrak{b} \subset \mathcal{O}_K : N\mathfrak{b} = l\}.$$

All ideals counted here are nonzero. For a sequence of fields and embedding orderings, we use the corresponding notation

$$D_i = |\text{Disc}(K_i)|, \quad Q_i = D_i^{1/2}, \quad \kappa_i = \text{Res}_{s=1} \zeta_{K_i}(s), \quad \mu_i = \mu_{K_i, \sigma_i}.$$

Unless another dependence is indicated, constants below are absolute for this fixed degree. In particular, they are independent of the field and of the ordering of its embeddings.

2.1 Volume and unfolding

We first make the sign convention in the packet definition explicit. The space $X = \text{SL}_6(\mathbb{Z}) \backslash \text{SL}_6(\mathbb{R})$ parametrizes ordinary unimodular lattices: a lattice has a positively oriented basis, and any two such bases differ by an element of $\text{SL}_6(\mathbb{Z})$. Consequently a diagonal sign matrix acts on lattices even when its determinant is -1 . The resulting lattice again represents a point of X by choosing a positively oriented basis. There is no additional orientation coordinate.

Let h_K and R_K denote the ordinary class number and regulator, respectively. Our convention for R_K is the covolume of the absolute logarithms of the units after deleting the sixth coordinate. Put

$$U = \mathcal{O}_K^\times, \quad U^+ = \{\epsilon \in U : \sigma_j(\epsilon) > 0 \ (1 \leq j \leq 6)\}, \quad s = [U : U^+].$$

For a nonnegative Borel function f on $\mathbb{R}^6$, define its lattice sum by

$$\hat{f}(\Lambda) = \sum_{v \in \Lambda \setminus \{0\}} f(v).$$

For $p > 0$, define a logarithmic fiber integral

$$V_f(p) = \sum_{\epsilon \in \{\pm 1\}^6} \int_{\mathbb{R}^5} f(\epsilon_1 e^{u_1}, \dots, \epsilon_5 e^{u_5}, \epsilon_6 p e^{-u_1 - \dots - u_5}) \, du_1 \cdots du_5. \quad (2.1)$$

Thus $V_f(p)$ integrates over the set on which the absolute product of the six coordinates is p , with the Haar normalization used for A .

Proposition 2.1 (Packet volume and vector unfolding). *Every orbit in $\mathcal{P}_{K, \sigma}$ has A -volume $(s/2)R_K$, and there are $h_K 2^6/s$ distinct orbits. In particular, the total packet volume is $Q\kappa$. For every nonnegative Borel function f ,*

$$\int_X \hat{f} \, d\mu_{K, \sigma} = \frac{1}{Q\kappa} \sum_{l \geq 1} d_K(l) V_f(l/Q), \quad (2.2)$$

where both sides are allowed to be infinite.

Proof. The only roots of unity in a totally real field are 1 and -1 . The kernel of the absolute logarithmic map on U therefore has order two. Its image on U^+ has index $s/2$ in its image on U . The unit theorem now gives covolume $(s/2)R_K$ for the positive-unit logarithms in the coordinates $u_1, \dots, u_5$.

We check that these are exactly the stabilizers and identifications in the packet. Suppose a real diagonal map sends $\sigma(I)$ onto $\sigma(J)$. Choose $0 \neq d \in I$, and let its image be $\sigma(e)$ with $e \in J$. All embeddings of d are nonzero, so the diagonal entries of the map are $\sigma_j(e/d)$. It is multiplication by $e/d \in K^\times$, and hence $J = (e/d)I$. This observation also applies to normalized lattices after their positive covolume factors have been undone. It follows that different ordinary ideal classes give disjoint packet orbits. For a fixed ideal, the diagonal automorphisms are its units, because $(I : I) = \mathcal{O}_K$.

The stabilizer in A is consequently the image of U^+ . Two sign translates belong to the same A -orbit exactly when their signs differ by a unit signature. The signature image has cardinality s , giving $2^6/s$ orbits for each ordinary class and the stated orbit volume. The analytic class number formula, in this normalization, is

$$Q\kappa = 2^5 h_K R_K.$$

It agrees with the product of the orbit number and orbit volume.

For completeness, let $\mathcal{F}$ be a fundamental domain for the positive-unit logarithms in the traceless diagonal algebra. Summing over ordinary ideal-class representatives and all sign matrices counts each orbit exactly s times. Thus for any nonnegative F on X ,

$$\int_X F d\mu_{K,\sigma} = \frac{1}{sQ\kappa} \sum_{[I]} \sum_w \int_{\mathcal{F}} F(\Lambda_{I,\sigma} w a(u)) du. \quad (2.3)$$

Here $du = du_1 \cdots du_5$ and the sixth coordinate is minus their sum.

Apply this identity to $F = \hat{f}$. A nonzero $d \in I$ determines the integral ideal

$$\mathfrak{b} = (d)I^{-1}, \quad \prod_{j=1}^6 \left| \text{covol}(\sigma(I))^{-1/6} \sigma_j(d) \right| = \frac{N\mathfrak{b}}{Q},$$

because $\text{covol}(\sigma(I)) = QNI$. The ideal $\mathfrak{b}$ determines the class $[I] = [\mathfrak{b}]^{-1}$, and every integral $\mathfrak{b}$ occurs. For that ideal, the possible d form a single U -orbit, or s orbits under U^+ . Summing a positive-unit orbit and integrating over $\mathcal{F}$ unfolds to the entire traceless diagonal algebra. Multiplication by a representative of U/U^+ translates the absolute logarithms and permutes the signs. After the sum over w , each of these s orbits therefore contributes $V_f(N\mathfrak{b}/Q)$. Their number cancels the factor s in (2.3), proving (2.2). All rearrangements are justified by nonnegativity. $\square$

2.2 Uniform counts of ideals

The normalization by κ in (2.2) makes it important to keep this same factor in upper bounds for ideal counts. We obtain it from a zero-free neighborhood of 1, followed by Shiu's estimate for nonnegative multiplicative functions.

Lemma 2.2 (Uniform ideal estimates). *For all sufficiently large discriminants D in the present family,*

$$\kappa \gg (\log D)^{-1}, \quad (2.4)$$

$$\sum_{p \leq D} \frac{d_K(p)}{p} \leq \log(\kappa \log D) + O(1), \quad (2.5)$$

$$\sum_{x-y < l \leq x} d_K(l) \ll y\kappa \quad \left(D^{1/4} \leq x \leq D, \ x^{1/2} \leq y \leq x \right). \quad (2.6)$$

In (2.5), p ranges over rational primes. The constants and the discriminant threshold are uniform in K .

Proof. We first record the precise zero exclusion needed here. Stark's Lemma 3 shows that the rectangle

$$\Re s \geq 1 - \frac{1}{4 \log D}, \quad |\Im s| \leq \frac{1}{4 \log D}$$

contains at most one nontrivial zero of ζ_K , and that such a zero is real and simple. His Lemma 8 applies also to nonnormal fields: a real zero β with

$$1 - \frac{1}{4 \cdot 6! \log D} \leq \beta < 1$$

would be a zero of the zeta function of a quadratic subfield of K [18, Lemmas 3 and 8]. Our field has no such subfield. Consequently, for an absolute $c_0 > 0$,

$$|1 - \rho| \geq \frac{c_0}{\log D} \quad \text{for every nontrivial zero } \rho \text{ of } \zeta_K. \quad (2.7)$$

One can take $c_0 = 1/2880$. This use of Stark's result does not require $K/\mathbb{Q}$ to be normal.

Set $s_0 = 1 + 1/\log D$. The logarithmic derivative of the completed Dedekind zeta function gives, for real $1 < s \leq s_0$,

$$S(s) := \sum_{\rho} \Re \frac{1}{s - \rho} = \frac{1}{s - 1} + \frac{1}{2} \log D + \frac{\zeta'_K(s)}{\zeta_K(s)} + O(1). \quad (2.8)$$

The zeros are counted with multiplicity. The real-part sum converges, and the gamma-factor contribution is bounded uniformly because the degree and signature are fixed. Each summand on the left is nonnegative, and $\zeta'_K(s_0)/\zeta_K(s_0) \leq 0$ by its Dirichlet series. Hence $S(s_0) \ll \log D$.

The same estimate holds throughout $(1, s_0]$. Indeed, writing $a = 1 - \Re \rho \geq 0$, $v = \Im \rho$, and $u = s - 1$, the relevant term is

$$\frac{a + u}{(a + u)^2 + v^2}.$$

For $0 < u \leq 1/\log D$, its numerator does not exceed its numerator at s_0 , while (2.7) gives $a^2 + v^2 \geq c_0^2/(\log D)^2$. The denominator at s_0 is therefore at most a constant multiple of the denominator at s . Termwise comparison proves the asserted uniform estimate for $S(s)$. Integrating (2.8) after moving $1/(s - 1)$ to the other side now gives

$$\zeta_K(s_0) \asymp \kappa \log D. \quad (2.9)$$

More explicitly, the logarithmic derivative of $(s - 1)\zeta_K(s)$ is $S(s) - \frac{1}{2} \log D + O(1)$, whose integral over this interval is $O(1)$, and its limit at $s = 1$ is κ . Since $\zeta_K(s_0) \geq 1$, this proves (2.4).

The Euler product gives

$$\sum_{p \leq D} \frac{d_K(p)}{p^{s_0}} \leq \log \zeta_K(s_0).$$

Also $d_K(p) \leq 6$ and

$$\sum_{p \leq D} d_K(p) \left(\frac{1}{p} - \frac{1}{p^{s_0}} \right) \leq \frac{6}{\log D} \sum_{p \leq D} \frac{\log p}{p} \ll 1,$$

where the final estimate follows from Chebyshev's bound by partial summation. Equation (2.9) now proves (2.5).

To obtain the interval estimate, we use Shiu's theorem [17], in the formulation of [15, Theorem 1.1] with empty excluded residue sets. For a nonnegative multiplicative function f , the hypotheses are

$$f(p^a) \leq A_1^a, \quad f(m) \leq A_2(\epsilon) m^\epsilon \quad (\epsilon > 0).$$

For any fixed $0 < \beta < 1/2$, the conclusion is

$$\sum_{x-y < m \leq x} f(m) \ll \frac{y}{\log x} \exp \left(\sum_{p \leq x} \frac{f(p)}{p} \right) \quad (x^\beta < y \leq x). \quad (2.10)$$

Both the constant and the sufficiently-large threshold for x depend only on β, A_1 , and the function A_2 .

The coefficients d_K are multiplicative and are bounded by the ordinary six-fold divisor function d_6 . In particular,

$$d_K(p^a) \leq \binom{a+5}{5} \leq 6^a, \quad d_K(m) \leq d_6(m) \ll_\epsilon m^\epsilon.$$

The hypotheses of (2.10) thus hold with constants independent of K . We take, for example, $\beta = 1/3$; the assumed length $y \geq x^{1/2}$ lies in the required range. Since $x \leq D$, (2.5) bounds the exponential factor by $O(\kappa \log D)$. Finally $x \geq D^{1/4}$ implies $\log D / \log x \leq 4$, proving (2.6). $\square$

2.3 The cusp and balls in the space of lattices

For a lattice $x \in X$, let $m(x)$ be the length of its shortest nonzero vector in the sup norm on $\mathbb{R}^6$. We use the sup norm only to simplify the fiber integral; replacing it by any fixed norm changes the constants.

Proposition 2.3 (Uniform cusp bound). *There are $C > 0$ and $c > 0$ such that, for all sufficiently large D ,*

$$\mu_{K,\sigma}\{x : m(x) \leq r\} \leq Cr^c \quad (0 < r < 1).$$

Consequently the packet probabilities form a tight family as $D \rightarrow \infty$, and every weak subsequential limit is an A -invariant probability measure on X .

Proof. Take $f = \mathbf{1}_{[-r,r]^6}$ and put $X_r = Qr^6$. For $0 < p \leq r^6$, the change of variables $t_j = \log r - u_j$ in (2.1) identifies the positive-sign integration region with

$$t_j \geq 0 \quad (1 \leq j \leq 5), \quad \sum_{j=1}^5 t_j \leq \log(r^6/p).$$

Thus

$$V_f(p) = \frac{2^6}{5!} (\log(r^6/p))^5 \quad (0 < p \leq r^6), \quad V_f(p) = 0 \quad (p > r^6).$$

Since the occurrence of a short vector is bounded by its count, (2.2) gives

$$\mu_{K,\sigma}\{m \leq r\} \ll \frac{1}{Q\kappa} \sum_{l \leq X_r} d_K(l)(1 + \log(X_r/l))^5. \quad (2.11)$$

The sum is empty if $X_r < 1$.

Suppose first that $r \geq D^{-1/48}$, so $X_r \geq D^{3/8}$. For $l \geq D^{1/4}$, divide the range of summation into intervals $(2^{-j-1}X_r, 2^{-j}X_r]$. Whenever such an interval contains one of these l , its upper endpoint is at least $D^{1/4}$ and at most Q . Lemma 2.2, with interval length half the upper endpoint, bounds its ideal count by $O(\kappa 2^{-j}X_r)$. Its weight in (2.11) is $O((j+1)^5)$. The total contribution of these terms before division by $Q\kappa$ is therefore at most

$$C\kappa X_r \sum_{j \geq 0} 2^{-j}(j+1)^5 \ll \kappa X_r.$$

For the remaining terms, the elementary divisor estimate $\sum_{l \leq Y} d_6(l) \ll Y(\log(2Y))^5$ gives an upper bound $CD^{1/4}(\log D)^{10}$ before normalization. In view of (2.4), its normalized contribution is at most

$$CD^{-1/4}(\log D)^{11} \ll r^6;$$

the last inequality follows from $r^6 \geq D^{-1/8}$. This proves an $O(r^6)$ bound in the present range.

Now let $r < D^{-1/48}$ and $X_r \geq 1$. Dyadic summation of the same elementary divisor estimate yields

$$\sum_{l \leq X_r} d_6(l)(1 + \log(X_r/l))^5 \ll X_r(\log(2X_r))^5.$$

Using $X_r \leq Q$ and (2.4), (2.11) is therefore at most $Cr^6(\log D)^6$. But $\log D < 48 \log(1/r)$ in this range, and $r^6(\log(1/r))^6 \ll r^3$. The claimed estimate follows, for example with $c = 3$, after increasing C .

Mahler's compactness criterion says that $\{x : m(x) \geq r\}$ is compact for every $r > 0$. The bound just proved therefore gives tightness. Any weak limit has mass one, and A -invariance passes to the limit because right translation by each fixed element of A is continuous and preserves compact supports. $\square$

Let $B(r)$ denote an open matrix ball about the identity in G , for r small; all fixed smooth local metrics give equivalent estimates.

Proposition 2.4 (Ordinary balls in a probability limit). *Suppose $D_i \rightarrow \infty$ and μ_i converges weakly to a probability μ . For every compact $\Omega \subset X$, there are constants $C_\Omega, r_\Omega > 0$ such that*

$$\mu(xB(r)) \leq C_\Omega r^6 \quad (x \in \Omega, 0 < r < r_\Omega).$$

Proof. At any lattice x_0 , choose a nonzero lattice vector none of whose coordinates is zero. Such a vector exists because a full-rank lattice cannot be contained in a finite union of proper linear subspaces. In an injective local chart at x_0 , continue this marked vector as $v(x)$. By reducing the chart, its coordinates remain bounded and bounded away from zero. A finite collection of these charts covers Ω , so these bounds and all local derivative bounds can be taken uniform.

For x in one of the charts and sufficiently small r , every lattice in $xB(r)$ has a nonzero vector in the ordinary Euclidean ball $B_{\mathbb{R}^6}(v(x), C_1 r)$. Let $f_{x,r}$ be the indicator of that ball. Its support has absolute coordinate product in an interval $J_{x,r}$ of length at most $C_2 r$; these intervals all lie

in a fixed compact subinterval of $(0, \infty)$ when $r < r_\Omega$. On this support the first five logarithmic coordinates each vary by $O(r)$. Consequently

$$V_{f_{x,r}}(p) \leq C_3 r^5 \mathbf{1}_{J_{x,r}}(p).$$

Fix $r > 0$. For all sufficiently large i , uniformly in the finitely many charts and in $x \in \Omega$, the ideal norms in $Q_i J_{x,r}$ can be covered by an interval with upper endpoint comparable to Q_i and length at most $C_4 Q_i r$. We may choose that length comparable to $Q_i r$, so it is at least the square root of its upper endpoint for large i . The upper endpoint lies between $D_i^{1/4}$ and D_i . Lemma 2.2 and (2.2) therefore give

$$\mu_i(xB(r)) \leq \int_X \widehat{f}_{x,r} d\mu_i \leq \frac{C_3 r^5}{Q_i \kappa_i} \sum_{l \in Q_i J_{x,r}} d_{K_i}(l) \leq C_5 r^6.$$

The constants are independent of x and r in the stated range; the threshold for i may depend on r . Since $xB(r)$ is open, the open-set inequality for weak convergence gives the same bound for $\mu(xB(r))$. This proves the assertion for each x and r with the uniform constants supplied by the finite cover. $\square$

2.4 Trace slices of the codifferent

The vector estimate has now supplied tightness and the ordinary-ball bound. To control diagonal tubes, we need to count a quantity which is nearly preserved by block-diagonal matrices. For an ordinary vector v and a dual vector y , that quantity is a partial sum of the products $v_j y_j$. The following lattice estimate makes such partial sums effective arithmetic coordinates.

Let $\mathfrak{D}_K$ be the different of K , so

$$\mathfrak{D}_K^{-1} = \{z \in K : \text{Tr}_{K/\mathbb{Q}}(z\mathcal{O}_K) \subset \mathbb{Z}\}.$$

Give $\mathbb{R}^6$ its Euclidean inner product, let $\mathbf{e} = (1, \dots, 1)$, and put

$$H = \mathbf{e}^\perp, \quad \mathcal{L}_K = \sigma(\mathfrak{D}_K^{-1}) \cap H.$$

Lemma 2.5 (Small cells in a trace slice). *The lattice $\mathcal{L}_K$ has Euclidean covolume $\sqrt{6}/Q$ in H and admits a fundamental paralleloptope of diameter $O(D^{-1/30})$. Let S be a nonempty proper subset of $\{1, \dots, 6\}$. For fixed $M > 0$, uniformly in $m \in \mathbb{Z}$ and intervals $J \subset \mathbb{R}$ of length $\delta > 0$,*

$$\# \left\{ z \in \mathfrak{D}_K^{-1} : \text{Tr}_{K/\mathbb{Q}} z = m, |\sigma_j(z)| \leq M \ (1 \leq j \leq 6), \sum_{j \in S} \sigma_j(z) \in J \right\} \ll_M Q(\delta + D^{-1/30}). \quad (2.12)$$

Proof. Since 1 is primitive in the free abelian group $\mathcal{O}_K$, choose an integral basis $\alpha_1 = 1, \alpha_2, \dots, \alpha_6$. Let $\beta_1, \dots, \beta_6$ be its trace-dual basis, so $\text{Tr}_{K/\mathbb{Q}}(\alpha_j \beta_k) = \delta_{jk}$. By the definition of the codifferent, these β_k form a basis of $\mathfrak{D}_K^{-1}$. In particular, $\text{Tr}_{K/\mathbb{Q}} \beta_k = \delta_{1k}$, which proves that the trace map onto $\mathbb{Z}$ is surjective and that $\sigma(\beta_2), \dots, \sigma(\beta_6)$ form a basis of $\mathcal{L}_K$. For $2 \leq j, k \leq 6$, orthogonal projection gives

$$\langle \text{proj}_H \sigma(\alpha_j), \sigma(\beta_k) \rangle = \delta_{jk}.$$

These projected vectors are therefore the dual basis inside H . Since $\text{proj}_H \sigma(\alpha_1) = 0$, we obtain

$$\mathcal{L}_K^* = \text{proj}_H \sigma(\mathcal{O}_K). \quad (2.13)$$

Take a nonzero vector in the right side of (2.13), arising from $\alpha \in \mathcal{O}_K$. The vector is nonzero precisely when $\alpha \notin \mathbb{Q}$. Primitivity of the field implies $\mathbb{Q}(\alpha) = K$, so the order $\mathbb{Z}[\alpha]$ has discriminant at least D . Differences of embedding coordinates are unchanged by orthogonal projection to H . It follows that

$$D \leq |\text{Disc}(\mathbb{Z}[\alpha])| = \prod_{j < k} |\sigma_j(\alpha) - \sigma_k(\alpha)|^2 \leq C \|\text{proj}_H \sigma(\alpha)\|^{30}.$$

Thus the first successive minimum of $\mathcal{L}_K^*$ is at least $cD^{1/30}$. Transference of successive minima in dimension five gives

$$\lambda_5(\mathcal{L}_K) \ll \lambda_1(\mathcal{L}_K^*)^{-1} \ll D^{-1/30}.$$

A fixed-dimensional lattice basis bound then provides a basis for $\mathcal{L}_K$ all of whose lengths are $O(D^{-1/30})$; these standard geometry-of-numbers facts may be found in [1]. The associated fundamental parallelotope has the asserted diameter.

The full codifferent lattice has covolume Q^{-1} . Its trace planes have spacing $1/\sqrt{6}$, since the trace takes every integer value and has normal vector $\mathbf{e}$. Decomposing a basis into five vectors in H and a vector of trace one therefore gives

$$Q^{-1} = \frac{1}{\sqrt{6}} \text{covol}_H(\mathcal{L}_K).$$

The trace- m points, if any, form a translate of $\mathcal{L}_K$. Tile that affine hyperplane by translates of the fundamental parallelotope, assigning boundaries measurably. A cell attached to a point counted in (2.12) lies in the $O(D^{-1/30})$ enlargement of its bounded slice. The linear functional $z \mapsto \sum_{j \in S} z_j$ is nonconstant on H because S is nonempty and proper. The five-dimensional volume of the enlarged slice is consequently $O_M(\delta + D^{-1/30})$, uniformly in its location and in m . Dividing by the cell covolume $\sqrt{6}/Q$ proves the counting bound. $\square$

2.5 Exponential decay on diagonal tubes

For D_i sufficiently large, define

$$T_{-,i} = \frac{\log D_i}{(\log \log D_i)^{2/3}}, \quad T_{+,i} = \frac{\log D_i}{(\log \log D_i)^{1/3}}. \quad (2.14)$$

Both tend to infinity, their ratio tends to infinity, and $T_{+,i} = o(\log D_i)$. The last property ensures that the cells in Lemma 2.5 are much smaller than the slices required at these times.

Proposition 2.6 (Two-sided tube estimate). *Let S be a nonempty proper subset of $\{1, \dots, 6\}$, and let $b = (b_1, \dots, b_6) \in L$ satisfy*

$$|b_j - b_k| \geq 1 \quad (j \in S, k \notin S).$$

For every compact $\Omega \subset X$, there is an identity neighborhood $O \subset G$ such that, for all sufficiently large i , all $x \in \Omega$, and all $T_{-,i} \leq t \leq T_{+,i}$,

$$\mu_i(x(a(tb)Oa(-tb) \cap a(-tb)Oa(tb))) \leq e^{-t/3}. \quad (2.15)$$

The neighborhood and discriminant threshold may be chosen simultaneously for any fixed finite collection of pairs (S, b) .

Proof. For a lattice Λ , its Euclidean dual is $\Lambda^* = \{y : \langle v, y \rangle \in \mathbb{Z} \text{ for all } v \in \Lambda\}$. At each point of Ω , choose nonzero marked vectors in the lattice and its dual, both avoiding all coordinate hyperplanes. Continue the pair in a local lattice chart. Finitely many such charts cover Ω , and the marked vectors in these charts lie in fixed bounded boxes separated from the coordinate hyperplanes. Their inner product is an integer, constant in each chart. Indeed, under right multiplication by h , the pair transforms as $v \mapsto vh$, $y \mapsto yh^{-\top}$.

Choose O sufficiently small in matrix coordinates. If

$$h \in a(tb)Oa(-tb) \cap a(-tb)Oa(tb),$$

then h is uniformly close to the identity, and each entry crossing the cut $S \sqcup S^c$ is $O(e^{-t})$. To verify the latter bound, the two membership conditions respectively bound its (j, k) entry by a constant times $e^{t(b_j - b_k)}$ and $e^{-t(b_j - b_k)}$; their minimum is $O(e^{-t|b_j - b_k|})$. Reducing O also makes h^{-1} uniformly bounded. Let P_S be the diagonal projection onto the coordinates in S . Then

$$hP_S h^{-1} - P_S = [h, P_S]h^{-1} = O(e^{-t}).$$

For the marked pair v_0, y_0 at the center x , put $v = v_0 h$ and $y = y_0 h^{-\top}$. These vectors remain in fixed bounded boxes avoiding the coordinate hyperplanes, and

$$\begin{aligned} \sum_{j=1}^6 v_j y_j &= \langle v_0, y_0 \rangle =: m \in \mathbb{Z}, \\ \sum_{j \in S} v_j y_j &= v_0 h P_S h^{-1} y_0^\top = v_0 P_S y_0^\top + O(e^{-t}). \end{aligned}$$

All constants are uniform over the finite cover of Ω . Thus membership in the tube forces the existence of an ordinary and dual vector pair satisfying these boundedness, trace, and partial-sum conditions. The short partial-sum interval may depend on x .

We count such pairs in the packet. For a representative ideal I and a packet lattice $c\sigma(I)wa$, where $c = \text{covol}(\sigma(I))^{-1/6}$, its dual is

$$c^{-1}\sigma(\mathfrak{D}_K^{-1}I^{-1})wa^{-1}.$$

Consequently a nonzero pair comes from

$$d \in I \setminus \{0\}, \quad \ell \in \mathfrak{D}_K^{-1}I^{-1} \setminus \{0\}.$$

The scalar, sign, and diagonal factors cancel in their coordinatewise product:

$$v_j y_j = \sigma_j(z), \quad z = d\ell \in \mathfrak{D}_K^{-1}. \quad (2.16)$$

The allowed z therefore belong to a fixed bounded embedding box, have trace m , and have their S -sum in an interval of length $O(e^{-t})$.

For a fixed nonzero allowed z , the ordinary vector's ideal $\mathfrak{b} = (d)I^{-1}$ must divide $(z)\mathfrak{D}_K$. In fact

$$\mathfrak{b}((\ell)I\mathfrak{D}_K) = (z)\mathfrak{D}_K,$$

and both factors on the left are integral. Conversely, this divisibility ensures $z/d \in \mathfrak{D}_K^{-1}I^{-1}$ whenever $(d)I^{-1} = \mathfrak{b}$. For a fixed d and z , there is exactly one possible ℓ , namely z/d .

Let f be the indicator of a fixed bounded box, or of a fixed finite union of boxes, containing all possible ordinary vectors v and separated from the coordinate hyperplanes. Its logarithmic fiber

integrals are bounded uniformly in $p > 0$, because the first five absolute logarithms range over a fixed bounded set. Repeating the unfolding of Proposition 2.1, with the additional restriction $\mathfrak{b} \mid (z)\mathfrak{D}_K$, gives

$$\int_X \#\{\text{admissible pairs with product } z\} d\mu_{K,\sigma} \leq \frac{1}{Q_K} \sum_{\mathfrak{b} \mid (z)\mathfrak{D}_K} V_f(N\mathfrak{b}/Q) \ll \frac{1}{Q_K} \#\{\mathfrak{b} : \mathfrak{b} \mid (z)\mathfrak{D}_K\}. \quad (2.17)$$

Here we have dropped the restrictions on the dual vector after fixing z , which can only increase the count. The divisibility restriction is unchanged by multiplication of d by a unit, so the same unfolding and sign multiplicities apply.

The embedding bound on z implies $N((z)\mathfrak{D}_K) = D|N_{K/\mathbb{Q}}z| \ll D$. For an integral ideal of norm N , its number of ideal divisors is at most $d(N)^6$, where d is the ordinary divisor function: over each rational prime there are at most six prime ideals, and every one of their exponents is at most the exponent of that prime in N . The classical divisor bound consequently gives, uniformly for our allowed z ,

$$\#\{\mathfrak{b} : \mathfrak{b} \mid (z)\mathfrak{D}_K\} \leq \exp\left(C \frac{\log D}{\log \log D}\right). \quad (2.18)$$

Lemma 2.5 bounds the number of allowed products by $O(Q(e^{-t} + D^{-1/30}))$. Uniformly for $t \leq T_{+,i}$,

$$D_i^{-1/30} = o(e^{-t}),$$

because $T_{+,i}/\log D_i \rightarrow 0$. The product count is therefore $O(Q_i e^{-t})$. The event defining the tube is bounded by the number of admissible pairs, so summing (2.17) over these products and using (2.18) gives

$$\mu_i(x(a(tb)Oa(-tb) \cap a(-tb)Oa(tb))) \ll \kappa_i^{-1} \exp\left(-t + C \frac{\log D_i}{\log \log D_i}\right).$$

Finally, (2.4) and $t \geq T_{-,i}$ show that the logarithm of the factor multiplying e^{-t} is $o(t)$, uniformly throughout the time range. Enlarging the discriminant threshold makes the last display at most $e^{-t/3}$. The same choices work for finitely many cuts and diagonal elements by taking the smallest of their identity neighborhoods and the largest of their thresholds. $\square$

3 Homogeneous limits and their fixed points

The estimates of Section 2 give probability limits and place a restriction on their local concentration. We first use this restriction to obtain positive entropy in almost every ergodic component. Measure classification then reduces the problem to excluding proper block orbits. The useful feature of those orbits is concrete: every point on one is fixed by a nonidentity diagonal matrix with algebraic entries.

3.1 Positive entropy in every positive-weight collection of components

For an invariant probability ν and an invertible measure-preserving map T , write $h_\nu(T)$ for its measure-theoretic entropy. Fix $b \in L$ with distinct coordinates and put

$$\lambda = \min_{j \neq k} |b_j - b_k| > 0, \quad T(x) = xa(b).$$

The next statement is the ordinary-ball argument of [13, Proposition 6.1 and Lemma 6.2], included here with its proof. It explains why a ball estimate for a measure can control its ergodic components without passing that estimate to each individual component.

Proposition 3.1. *Let μ be an A -invariant probability on X . Suppose that, for every compact $\Omega \subset X$, there are $C_\Omega, r_\Omega > 0$ such that*

$$\mu(xB(r)) \leq C_\Omega r^n \quad (x \in \Omega, \ 0 < r < r_\Omega), \quad (3.1)$$

where $B(r)$ is a matrix neighborhood of the identity of radius r . Every A -invariant probability $\xi \leq c\mu$, with $c < \infty$, satisfies $h_\xi(T) \geq \lambda/3$. Consequently almost every A -ergodic component of μ has positive entropy for T .

Proof. Choose a sufficiently small fixed $\epsilon > 0$ and consider the two-sided tube

$$\mathcal{T}_t = a(tb)B(\epsilon)a(-tb) \cap a(-tb)B(\epsilon)a(tb).$$

If $h \in \mathcal{T}_t$ and $r = e^{-\lambda t}$, then

$$|h_{jj} - 1| < \epsilon, \quad |h_{jk}| \leq \epsilon r \quad (j \neq k).$$

Every nondiagonal term in the determinant expansion contains at least two off-diagonal entries. Hence $\prod_j h_{jj} = 1 + O(r^2)$. The matrix

$$d(h) = \text{diag} \left(h_{11}, \dots, h_{n-1, n-1}, \left(\prod_{j < n} h_{jj} \right)^{-1} \right)$$

belongs to a fixed compact set $\mathcal{D} \subset A$ and satisfies $d(h)^{-1}h \in B(Cr)$. An r -mesh of the $(n-1)$ -dimensional set $\mathcal{D}$ therefore covers $\mathcal{T}_t$ by at most $Cr^{-(n-1)}$ sets $d_j B(C'r)$, with $d_j \in \mathcal{D}$. For $x \in \Omega$, all centers xd_j belong to the compact set $\Omega\mathcal{D}$. Domination and (3.1) give

$$\xi(x\mathcal{T}_t) \leq C_{\Omega, c} e^{-\lambda t}. \quad (3.2)$$

We apply the entropy criterion of Einsiedler–Lindenstrauss–Michel–Venkatesh [8, Corollary 3.3]. In the form needed here, it says that if probabilities ν_i invariant under $a(\mathbb{R}b)$ converge to a probability ν , and $t_i \rightarrow \infty$, then

$$\nu_i(x[a(t_i b)Oa(-t_i b) \cap a(-t_i b)Oa(t_i b)]) \leq C_\Omega e^{-2\eta t_i} \quad (x \in \Omega)$$

for every compact Ω , with an identity neighborhood O allowed to depend on Ω , implies $h_\nu(T) \geq \eta$. This criterion includes the noncompact space X and requires that the limit have total mass one. Take the constant sequence $\nu_i = \xi$, $t_i = i$, and $\eta = \lambda/3$. Equation (3.2) verifies its hypothesis.

Write the A -ergodic decomposition as $\mu = \int_Z \nu_z \, d\tau(z)$. The function $z \mapsto h_{\nu_z}(T)$ is measurable: for increasing finite Borel partitions $\mathcal{P}_k$ generating the Borel sets, it is given by

$$h_{\nu_z}(T) = \sup_k \inf_{N \geq 1} \frac{1}{N} H_{\nu_z} \left(\bigvee_{j=0}^{N-1} T^{-j} \mathcal{P}_k \right).$$

The entropy integral formula [16, Theorem 9.8] applies to this decomposition into T -invariant probabilities. The component partition is fixed by T , so the transformation on its quotient is the identity and has zero entropy. The formula remains valid after restricting to a measurable collection of components; the components need not themselves be T -ergodic.

If the zero-entropy components had weight $w > 0$, their normalized average ξ would satisfy $\xi \leq w^{-1}\mu$ and $h_\xi(T) = 0$. This contradicts the first part of the proposition. $\square$

3.2 The algebraic alternatives

We use the following measure-classification theorem in its full-degree form.

Theorem 3.2 (Einsiedler–Katok–Lindenstrauss). *For $n \geq 3$, an A -invariant, A -ergodic probability on $\mathrm{SL}_n(\mathbb{Z}) \backslash \mathrm{SL}_n(\mathbb{R})$ that has positive entropy for some diagonal one-parameter subgroup is the invariant probability on a closed finite-volume orbit xJ , where J is a closed connected subgroup containing A .*

This is [7, Theorem 1.3], with the definition of an algebraic measure given there. Their left-action convention is transferred to ours by $\Gamma g \mapsto g^{-1}\Gamma$; this replaces $a(b)$ by $a(-b)$, which has the same entropy. The Haar conclusion for prime n in [7, Corollary 1.4] does not identify the sextic alternatives. We shall distinguish them using their fixed points.

Lemma 3.3. *Let J be a connected closed subgroup of G containing A , and suppose xJ carries a finite nonzero J -invariant measure. There is a partition $\{1, \dots, n\} = B_1 \sqcup \dots \sqcup B_d$ such that J is the identity component of the determinant-one block-diagonal group for this partition.*

Proof. The Lie algebra $\mathfrak{j}$ contains L and is invariant under $\mathrm{Ad}(A)$. Since the off-diagonal weight spaces are one-dimensional, $\mathfrak{j}$ is the direct sum of L and a collection of root lines $\mathbb{R}E_{jk}$. Draw the edge $j \rightarrow k$ when this line occurs. The bracket $[E_{ij}, E_{jk}] = E_{ik}$ makes the graph transitively closed whenever $i \neq k$.

A Lie group with a lattice is unimodular. Applying the vanishing of its modular character to A shows that

$$\sum_{j \rightarrow k} (b_j - b_k) = 0 \quad (b \in L).$$

Thus each vertex has equal indegree and outdegree. Every strongly connected class is complete by transitivity. The graph of these classes is acyclic. If it had an edge, a source class of an edge-containing component would have strictly more outgoing than incoming edges in the original graph, contradicting the sum of the degree balances over that class. Hence there are no edges between classes. The resulting Lie algebra is precisely the block-diagonal one. A connected Lie subgroup is determined by its Lie algebra, giving the assertion. $\square$

The next lemma supplies a countable obstruction to each proper orbit. It uses finite volume twice: first to identify a rational commutant and then to exclude its idempotents.

Lemma 3.4. *Every proper orbit xJ in Theorem 3.2 is contained in $\mathrm{Fix}(a) = \{y \in X : ya = y\}$ for some nonidentity $a \in A$ whose entries are algebraic numbers. There are only countably many possible matrices a . For each such a , its fixed set is covered by countably many smooth pieces. Whenever $a_{jj} \neq a_{kk}$, the direction of $U_{jk} = \{1 + tE_{jk} : t \in \mathbb{R}\}$ is nowhere tangent to these pieces, and each piece meets a sufficiently small U_{jk} plaque in at most one point.*

Proof. Write $x = \Gamma g$ and $\Delta = g^{-1}\Gamma g \cap J$. This is a lattice in J . For the rational linear algebra in this proof, use the auxiliary column lattice $\Lambda = g^{-1}\mathbb{Z}^n$, which Δ preserves, and its rational span $\Lambda_{\mathbb{Q}}$. This convention is separate from the row-lattice realization of the point x . We first show that the real matrix commutant of Δ equals the commutant of J . For any matrix C commuting with Δ , the map

$$\Delta \backslash J \longrightarrow \mathrm{Mat}_n(\mathbb{R}), \quad \Delta j \longmapsto j^{-1}Cj$$

is well-defined. It sends normalized Haar measure to a probability invariant under conjugation by A . Choose a regular element of A . Every off-diagonal matrix coordinate has a nonzero weight under its

conjugation. Recurrence of an invariant probability forces all such coordinates to vanish almost surely: a nonzero coordinate expands geometrically in one of the two time directions and cannot recur to a bounded set. The map is continuous and Haar measure has full support, so $j^{-1}Cj$ is diagonal for every $j \in J$.

Taking $j = 1$ shows that C is diagonal. Conjugating by a root subgroup within one of the blocks of Lemma 3.3 forces the two corresponding diagonal entries of C to agree. Consequently the commutant of Δ is exactly the algebra of block scalars, namely the commutant of J .

Define

$$F = \{C \in \text{End}_{\mathbb{Q}}(\Lambda_{\mathbb{Q}}) : C\delta = \delta C \text{ for all } \delta \in \Delta\}.$$

In the lattice coordinates supplied by g , these commutation relations are rational linear equations, since $g\Delta g^{-1} \subset \Gamma$. Only finitely many of these equations are needed in the finite-dimensional matrix space. Thus F has real extension isomorphic to $\mathbb{R}^d$, where d is the number of blocks. In particular F is a commutative, reduced, finite-dimensional $\mathbb{Q}$ -algebra, a product of totally real fields.

Suppose F had a nontrivial idempotent. Its image in real coordinates would select a nonempty proper union of blocks and hence a subspace V defined over the rational lattice. The group Δ preserves the full lattice in V and its inverse does too. Thus

$$\chi(j) = |\det(j|_V)|$$

is a character of J trivial on Δ . It is nontrivial on A , because both the selected and the unselected union of blocks are nonempty. It is therefore onto $\mathbb{R}_{>0}$. Pushing the finite invariant measure on $\Delta \backslash J$ forward by χ would give a finite nonzero multiplicatively invariant measure on $\mathbb{R}_{>0}$, an impossibility. Hence F is a field.

The representation of F on the rational lattice space has some dimension m over F . After extending scalars to $\mathbb{R}$, each embedding of F has multiplicity m , so all blocks have size m and $dm = n$. The elements of F carrying Λ into itself form an order. Indeed, in lattice coordinates this multiplier set is represented by $(gFg^{-1}) \cap \text{Mat}_n(\mathbb{Z})$, hence is discrete. Clearing the denominators of a rational basis gives a full-rank submodule. Thus the multiplier set is a finitely generated group of rank d , and it is a subring containing 1. Dirichlet's unit theorem for this order gives unit rank $d - 1$.

If $J \neq G$, then $d > 1$. Choose a unit of infinite order and take an even power so that every real embedding is positive. Its action on the lattice is an automorphism with determinant $N_{F/\mathbb{Q}}(u)^m = 1$. In real coordinates it is a matrix $a \in A$, constant on each block and therefore central in J . It fixes x , and centrality gives $(xj)a = (xa)j = xj$ for every $j \in J$. Its entries are conjugates of an algebraic unit. The collection of possible such diagonal matrices is countable.

Finally, fixing a and lifting to G , the fixed-point equation is

$$gag^{-1} = \gamma \quad \text{for some } \gamma \in \Gamma.$$

For each γ , the nonempty solution set is a coset of the centralizer of a . It is smooth, with tangent root directions exactly those satisfying $a_{jj} = a_{kk}$. Restricting to countably many injective local quotient charts gives the asserted pieces in X . If $a_{jj} \neq a_{kk}$, two points of one local U_{jk} plaque cannot solve the same lifted equation: their relative element $1 + tE_{jk}$ would commute with a , forcing $t = 0$. This proves the last assertion. $\square$

For $n = 6$, the equality $dm = 6$ leaves only block sizes $3 + 3$, $2 + 2 + 2$, or $1 + 1 + 1 + 1 + 1 + 1$ for a proper finite-volume orbit. The last case is $J = A$, whose translation action on a compact quotient has zero entropy. Thus the positive-entropy alternatives that must be excluded have two blocks of size three or three blocks of size two.

By Proposition 2.4, Proposition 3.1, and Theorem 3.2, almost every ergodic component of any packet limit is one of these homogeneous probabilities. To complete the proof, it will suffice to show that such a limit assigns no mass to the countable fixed-point pieces in Lemma 3.4. We now construct conditional measures that can detect those pieces while retaining the arithmetic information at the growing times of Proposition 2.6.

4 Conditional probabilities at intermediate resolutions

We now pass from the measures μ_i to conditional probabilities along unipotent plaques. The conditioning must be performed before taking the limit: exact conditional measures of a compact diagonal orbit along a unipotent group contain no information about its distribution at the scales of the arithmetic estimate. We instead condition on small *transverse cells*. The freedom to choose their depth over a broad range will make the resulting data compatible with changes of charts and with certain diagonal motions whose length tends to infinity.

Throughout this section, $G = \mathrm{SL}_n(\mathbb{R})$, $X = \mathrm{SL}_n(\mathbb{Z}) \backslash G$, and $n = 6$ in the application. We assume only that μ_i are A -invariant probability measures converging weakly to an A -invariant probability μ , and that the available times satisfy

$$T_{-,i} \longrightarrow \infty, \quad T_{+,i}/T_{-,i} \longrightarrow \infty.$$

The particular arithmetic values of these times will be used later. All subsequences taken below preserve these assumptions.

4.1 Plaque charts and their extensions

Write L for the traceless diagonal algebra and $a(b) = \exp(b)$ for $b \in L$. For $j \neq k$, set

$$\beta_{jk}(b) = b_k - b_j, \quad U_{jk} = \{1 + tE_{jk} : t \in \mathbb{R}\}.$$

These signs correspond to the right action on X : the displacement u at x becomes $a(-b)ua(b)$ at $xa(b)$, and its jk parameter is multiplied by $e^{\beta_{jk}(b)}$.

A *pattern* is a set of roots positive on some element of L and closed under root addition whenever the sum is a root. Its *pattern group* V consists of the unipotent matrices with arbitrary entries in those positions and zero entries elsewhere. After permuting coordinates it is a triangular unipotent group. All these groups are normalized by A , and there are only finitely many of them. We also use the groups VA , but only in local charts.

For any such group H , a plaque chart is an injective smooth map

$$\Phi : Q \times B \longrightarrow X, \quad \Phi(q, h) = \psi(q)h, \tag{4.1}$$

where Q is an open box in a complementary transverse manifold and B is an open bounded coordinate box in H . The map extends smoothly and injectively to a slightly larger domain. The sets with q fixed are its plaques. Smooth reparametrizations of h , allowed to depend smoothly on q , describe the same plaques and will also be used. A smaller chart with closure in the displayed chart supplies a fixed margin from its boundary.

The comparison of probabilities on different bounded portions of a plaque requires charts extending beyond a prescribed displacement range. The following elementary recurrence observation supplies them.

Lemma 4.1 (No periods in a pattern group). *Let ν be an A -invariant probability on X and let V be a pattern group. For ν -almost every x ,*

$$xv = x, \quad v \in V \quad \implies \quad v = 1.$$

At every such x , the image xK of any prescribed compact displacement range $K \subset V$ is contained in an embedded plaque with a transverse neighborhood. Consequently, for any compact displacement range in V and any $\epsilon > 0$, finitely many inner plaque charts cover a set of ν -measure at least $1 - \epsilon$, and their outer charts contain that range about every point of the corresponding inner charts.

Proof. Choose $b \in L$ such that $a(-tb)va(tb) \rightarrow 1$ as $t \rightarrow \infty$ for every $v \in V$. At a point recurrent under $a(b)$, a relation $xv = x$ gives

$$xa(kb)a(-kb)va(kb) = xa(kb).$$

A subsequence of these points returns to a compact neighborhood of x . Stabilizers of points in that neighborhood contain no nonidentity element in a fixed neighborhood of 1: this follows from the discreteness of the lattice and local quotient charts. The displayed relation is therefore impossible for $v \neq 1$. Poincaré recurrence applies at almost every point.

The orbit map $v \mapsto xv$ is then an injective immersion. Its restriction to a compact set is an embedding. Take a slightly larger compact coordinate domain in V and use a complementary transverse slice. Compactness permits shrinking this slice until its product with the prescribed plaque domain is injective. Indeed, a failure along slices shrinking to x would give either two distinct points of the limiting embedded plaque with the same image, or contradict local injectivity near one of its points. Shrinking once more provides inner charts with margins. A countable subcover and continuity from below give the finite-cover assertion. $\square$

Fix a countable stock of plaque charts and inner charts with the following properties. It is a basis wherever local charts are needed; for pattern groups it also supplies the extensions in Lemma 4.1, for each compact displacement range in a countable exhaustion. It includes subboxes, fixed subdivisions of the plaque coordinates into arbitrarily fine bins, and the simultaneous coordinate usages described next. Chart and bin boundaries have μ -measure zero. Such choices are possible by varying coordinate endpoints; for each coordinate only countably many levels can have positive measure. One may start with multiplication charts and choose countably many centers, transversals, and domain sizes dense among the choices with compact closure and an injective enlargement.

Two simultaneous usages will matter. If $V = WU$, where U is a root group, W is a normal pattern subgroup, and the patterns are disjoint, include charts

$$(q, w, u) \mapsto \psi(q)wu.$$

Their V , U , and W transverse coordinates are, respectively, q , (q, w) , and (q, u) . Normality gives the last assertion; the w coordinate is a smoothly reparametrized W coordinate when u is fixed. Also include charts $(q, v, a) \mapsto \psi(q)va$ for VA , with the simultaneous V transverse coordinate (q, a) . All chart bounds, plaque extensions, bin sizes, and fixed margins in a comparison are chosen before i tends to infinity.

4.2 A broad range of almost unchanged conditionals

In a chart $\mathcal{B} = \Phi(Q \times B)$, restrict μ_i to $\mathcal{B}$. Partition Q by a shifted dyadic grid of side 2^{-l} . For a point $x = \Phi(q, h)$ in the chart, let

$$P_{i,l}^{\mathcal{B}}(x) = \text{the conditional distribution of } h \text{ given the grid cell of } q.$$

The conditioning uses the restricted measure. Zero-mass columns may be assigned any probability. An expectation involving this chart means integration over $\mathcal{B}$ with the unnormalized measure $\mu_i|_{\mathcal{B}}$; equivalently, one can normalize that measure and then multiply the expectation by $\mu_i(\mathcal{B})$.

For positive sequences r_i, s_i , write $r \prec s$ if $r_i/s_i \rightarrow 0$. An interior depth will mean a sequence d_i with $l_- \prec d \prec l_+$, where the endpoints are supplied by the next lemma. Indices i on scales will often be suppressed.

Lemma 4.2 (A common plateau). *After passage to a subsequence there are integer depths $l_{-,i}$ and $l_{+,i}$ such that*

$$T_- \prec l_- \prec l_+ \prec T_+. \quad (4.2)$$

For every chart and every fixed finite plaque-bin partition in the countable stock, the expected ℓ^1 distance between the conditional bin-probability vectors at depths l_- and l_+ tends to zero, averaged also over a uniform shift of the nested dyadic grid. The same conclusion holds between any two intervening depths for that grid.

The shifts can be selected so that these conclusions hold for every fixed chart and fixed bin partition and so that random-grid boundary estimates hold at the two endpoint depths and at any prescribed countable collection of interior depth sequences. When coordinate usages share transverse labels, their grids may share the corresponding shifts. At interior depths, conditional bin probabilities are unchanged in expected ℓ^1 distance by using auxiliary shifted grids selected with the same boundary convention.

Proof. Fix a chart, a finite bin partition, and a grid shift. If p_l is the conditional bin-probability vector, its coordinates form bounded martingales in the increasing transverse grid sigma-fields. Thus

$$\int_{\mathcal{B}} \|p_v - p_u\|_2^2 d\mu_i = \int_{\mathcal{B}} \|p_v\|_2^2 d\mu_i - \int_{\mathcal{B}} \|p_u\|_2^2 d\mu_i, \quad u \leq v.$$

The sum of these differences over disjoint depth intervals is at most 1. This remains true after averaging in the shift. Enumerate all fixed chart/bin choices and assign them positive summable weights. Their weighted sum has the same bounded total variation.

Put $L_i = \log(T_{+,i}/T_{-,i})$. Divide the interval of logarithmic depths into m_i consecutive intervals, where $m_i \rightarrow \infty$ and $m_i = o(L_i)$. Discard its first and last intervals. Some remaining interval has weighted martingale cost $O(m_i^{-1})$. Its endpoints, rounded to integers, give l_- and l_+ . The discarded end intervals and the length $L_i/m_i \rightarrow \infty$ give all three ratios in (4.2). Each fixed chart/bin choice has cost tending to zero; Cauchy–Schwarz gives the asserted ℓ^1 conclusion. Orthogonality of martingale increments gives it for intervening depths as well.

For the boundary assertion, fix a point q in a Euclidean transverse coordinate domain. Under a uniform grid shift, the probability that q is within $\rho 2^{-l}$ of a depth- l grid wall is $O(\rho)$, uniformly in q, l , and the sampling measure. Integrating first in the shift gives the same estimate for any measure on that domain. Choose $\rho_i \downarrow 0$ slowly. Markov’s inequality, followed by a diagonal choice over the chart/bin and depth lists, selects shifts with both the conditional-probability estimates and boundary masses tending to zero. Common labels can use common shift variables: each marginal shift remains uniform, which is all the argument uses.

Here is the comparison with an auxiliary grid at an interior depth d . Outside negligible boundary mass, one of its cells lies inside one depth- l_- cell and is a union of depth- l_+ cells, apart from the latter cells meeting its walls. The relevant relative collar widths are $O(2^{l_- - d})$ and $O(2^{d - l_+})$. Both tend to zero much faster than the chosen ρ_i . Summing the martingale bin errors over the contained fine cells compares the auxiliary conditional law to the coarse one. The discarded mass changes the weighted conditional ℓ^1 error by at most twice that mass. This proves the last assertion. The

same argument compares any two of the selected grids by sandwiching them between these endpoint grids. $\square$

We make the selection convention precise for later use. All defining charts, simultaneous coordinate usages, fixed bin subdivisions, and depth sequences belong to countable lists. A diagonal selection enforces their estimates for each fixed member; it does not assert uniformity over the whole infinite list at a finite index i . Within $[l_-, l_+]$ any fixed finite number of scales can be chosen with successive ratios tending to infinity and with room at both ends. For instance, use distinct fixed-fraction interpolations on the logarithmic depth interval. Different arguments may reuse such scales. Auxiliary grids chosen for one comparison need not define the data, because Lemma 4.2 compares their fixed-bin laws with those of the defining grids. Such a grid may also be chosen later at an interior depth not in the original list; it does not change the augmented states defined below. We always let ρ_i decrease slowly enough to dominate the exponentially small relative collar widths arising from these scale separations. None of these conventions requires absolute continuity.

4.3 The augmented probability space

Choose a defining interior depth for every chart and selected grids as above. Record at x its conditional plaque probability in each chart containing it, and an isolated dummy symbol for every other chart. Write $Z_i(x)$ for the resulting tuple together with x . Each plaque probability is regarded as a probability on the compact closure of its coordinate domain. The countable product of these compact spaces, together with the X coordinate, is a separable metrizable space. Tightness of μ_i therefore permits passage to a subsequence for which the laws of $Z_i(x)$ converge. Denote the limiting state by

$$\omega = (x, (P^\mathcal{B})_\mathcal{B}).$$

Its x marginal is μ . Fixed-bin probabilities pass to the limit: the averaged plaque probability in a chart is its plaque-coordinate marginal, so the null-boundary choices imply that the limiting probabilities assign zero mass to the chosen bin boundaries almost surely.

Lemma 4.3 (Sampling one plaque probability). *In any fixed chart, conditional on its transverse coordinate q and its own limiting probability $P^\mathcal{B}$, the sampled plaque coordinate h has law $P^\mathcal{B}$. In particular, $h \in \text{supp } P^\mathcal{B}$ almost surely on that chart. This assertion conditions on this one probability, rather than on the entire augmented state.*

Proof. Let q_i^0 be a representative of the cell containing q . At the sequence level, $P_i^\mathcal{B}$ is a function of that cell. Consequently, for bounded continuous tests f and F ,

$$\int_{\mathcal{B}} f(h) F(q_i^0, P_i^\mathcal{B}) d\mu_i = \int_{\mathcal{B}} P_i^\mathcal{B}(f) F(q_i^0, P_i^\mathcal{B}) d\mu_i.$$

On compact inner boxes, replacing q_i^0 by q incurs an error tending to zero. Let $i \rightarrow \infty$ and then exhaust the chart by inner boxes. Its boundary has zero limiting mass. This proves the conditional-law identity. The standard support assertion for a sample from a probability measure follows, for example, by using a countable basis of the plaque domain. $\square$

The remaining task in this section is to show that probabilities recorded in different charts describe the same measure along a plaque, up to normalization. We first isolate the estimate that allows many small transverse columns to be regrouped without losing control of the error.

4.4 Aggregating fine columns into patches

In an output chart use the defining nested grid at depths l_- and l_+ . Write E for a coarse cell and e for a fine cell contained in E . Fix finitely many plaque bins J_j , and set

$$M_e = \mu_i(e \text{ column}), \quad M_{e,j} = \mu_i(e \text{ column with } h \in J_j), \quad P_E(j) = \mathbb{P}_i(h \in J_j \mid q \in E).$$

Here and below the chart restriction is understood. The plateau estimate is precisely

$$\delta_i := \sum_E \sum_{e \subset E} \sum_j |M_{e,j} - M_e P_E(j)| \longrightarrow 0. \quad (4.3)$$

Lemma 4.4 (Comparison on trimmed patches). *In the preceding chart, let p range over pairwise disjoint measurable patches, each lying transversely in one coarse cell $E(p)$. Suppose a subset of each patch has the product form*

$$p^{\text{in}} = \bigcup_{e \in I_p} (e \times B_p), \quad B_p = \bigcup_{j \in J(p)} J_j,$$

and the sum of the discarded masses $\mu_i(p \setminus p^{\text{in}})$ is at most ϵ . The collections $I_p, J(p)$ may depend on i, p , and the number of patches need not be bounded. If L_p is the plaque-bin law conditional on p , then, with errors weighted by $\mu_i(p)$,

$$L_p \text{ agrees with } \frac{\mathbf{1}_{B_p} P_{E(p)}}{P_{E(p)}(B_p)} \text{ to total expected } \ell^1 \text{ error } O(\epsilon) + o(1).$$

An arbitrary law is used if the denominator vanishes. The unnormalized version is

$$\sum_p \mu_i(p) \left\| P_{E(p)}(B_p) L_p - \mathbf{1}_{B_p} P_{E(p)} \right\|_1 \leq O(\epsilon) + o(1). \quad (4.4)$$

If patches and a disjoint refinement into subpatches admit such trims with the same coarse cell and the same plaque-bin union, their conditional bin laws agree with one another with the corresponding weighted error.

These assertions also hold for bounded continuous tests, up to their uniform continuity error on the fixed bins. A test may depend on the sampled point, provided its norm and continuity modulus in the tested plaque variable are uniform. At the output sample, $P_{E(p)}$ may be replaced in fixed-bin expressions by the recorded data probability, with an additional error tending to zero.

Proof. For a trimmed patch, sum the errors in (4.3) over $e \in I_p$ and $j \in J(p)$. The resulting comparison is between its unnormalized bin-mass vector and

$$\left(\sum_{e \in I_p} M_e \right) \mathbf{1}_{B_p} P_{E(p)}.$$

Every pair (e, j) occurs in at most one trimmed patch, since those sets are disjoint. Hence the total unnormalized error is at most δ_i . Restoring discarded mass adds at most ϵ . For two nonnegative finite vectors v, w , comparison after normalization, weighted by $\|v\|_1$, costs at most $2\|v - w\|_1$; this follows by adding and subtracting $w/\|v\|_1$. The same inequality covers zero mass by an arbitrary choice of normalized vector. It proves the first assertion and, on multiplying by $P_{E(p)}(B_p) \leq 1$, proves (4.4).

Apply the same argument separately to patches and subpatches and use their common restricted vector $P_E|_{B_p}$ to obtain their comparison. Approximating a continuous test by its values on bins gives the remaining test assertion. The bound is in ℓ^1 , so it is uniform over all choices of bounded bin values, including values depending on the sample. Finally, the plateau controls the expected fixed-bin distance between the coarse conditional vector and the recorded vector. Restricting to disjoint patches does not increase that expectation. $\square$

In applications the bins and charts are fixed first. Their sizes may then be chosen sufficiently small to achieve any prescribed limsup error, after which i tends to infinity. Thus “arbitrarily small error” below means an actual bound by a prescribed $\epsilon > 0$ in this order of choices. It is not a claim about uniform convergence over all subdivisions.

4.5 Projective agreement and diagonal motions

Center the probability in a V chart at its sampled point: if the sample has group coordinate h_s , the group displacement of h is $h_s^{-1}h$. For a smoothly reparametrized chart, use the corresponding group elements in this formula. Two nonzero measures on a common domain agree *projectively* if one is a positive scalar multiple of the other there.

Proposition 4.5 (Comparison of plaque data). *For the augmented limit constructed above the following assertions hold.*

1. *For every pattern group V , its centered probabilities in different charts agree projectively on their common plaque sheet near the sampled point. They extend to a measurable nonzero projective locally finite measure m_ω^V on V , with $1 \in \text{supp } m_\omega^V$. The chart probabilities represent its restrictions on all bounded displacement ranges contained in their plaque domains.*
2. *Fix $b \in L$. In every subsequential joint limit of*

$$(Z_i(x), Z_i(xa(b))), \quad x \sim \mu_i,$$

the two V measures are related projectively by the pushforward under $v \mapsto a(-b)va(b)$.

3. *Fix a root β and an interior time scale R with room for interior depths on both sides. For every fixed $C < \infty$ and every sequence*

$$b_i \in \ker \beta, \quad \|b_i\| \leq CR_i,$$

the measures m^{U_β} in every subsequential joint limit of $(Z_i(x), Z_i(xa(b_i)))$ agree projectively. This assertion holds for every such sequence b_i ; the defining grids and state construction are unchanged.

Proof. We prove one local comparison that covers all three assertions. The motion is either the identity, a fixed $a(b)$, or $a(b_i)$ with $b_i \in \ker \beta$ and $V = U_\beta$. Denote its induced conjugation on V by c . In the first two cases c is fixed; in the third case c is the identity.

We display the geometry in group-element plaque coordinates. For a reparametrized chart, write its original coordinate as $t \in B_0$ and its group element as $h = r_q(t)$, where the maps r_q and their inverses are smooth on the fixed extended chart. Its group-coordinate domain is $B(q) = r_q(B_0)$; for a multiplication chart $B(q)$ is constant. The conditional probabilities and bins remain in the original fixed coordinates. When converting a probability to group coordinates at a sample, use r_{q_s} . Across a depth- F cell this conversion differs from the actual r_q by $O(2^{-F})$, which has vanishing effect on every continuous test considered below.

In the long-motion case choose an input transverse depth F with

$$l_- \prec R \prec F \prec l_+, \quad (4.5)$$

For a fixed motion, it suffices to take $l_- \prec F \prec l_+$; all factors $\exp(O(R))$ below are then bounded constants. The depth- F input grid may be an auxiliary grid selected for this comparison. Thus R and F need not belong to the countable list used to define the states; Lemma 4.2 compares the auxiliary laws back to the recorded ones. Its boundary selection uses μ_i and the scale bound, and does not depend on the particular b_i within that bound. The input patches are the depth- F transverse cells times the full plaque-coordinate domain. Their conditional laws are the input probabilities at depth F .

Fitting the images into output charts. Choose input inner charts and fixed displacement test ranges first. An input chart may have any fixed plaque extension needed for these ranges. By Lemma 4.1, output inner charts can be chosen whose outer charts accommodate the entire image plaque range, with a margin. A finite such cover omits arbitrarily small limiting mass. The output distribution is again μ_i , so the same is true for the sequence. If an image patch meets one of these inner charts, its whole image fits in the corresponding outer chart for all sufficiently large i . This last assertion follows from the transverse diameter estimate below. Assign such patches to output charts, or work with each applicable pair of charts; the finite number of pairs only multiplies error bounds by a fixed constant.

More precisely, let $B(q)$ and $B'(q')$ be the full input and output domains in group coordinates, and let $(q_s, h_s), (q'_s, h'_s)$ be the two samples. The output extension is required to contain

$$h'_s c(h_s^{-1} \overline{B(q_s)}) \subset B'(q'_s) \quad \text{with a positive boundary margin.} \quad (4.6)$$

This condition concerns the entire input plaque domain. It is an open condition on the paired sample coordinates: the domains are smooth images of fixed compact closures. Their possible centered ranges lie in a fixed compact subset of V , so the extension lemma supplies output covers with this property.

In the two charts the map has the form

$$(q, h) \mapsto (\theta_i(q), d_i(q)c(h)). \quad (4.7)$$

The first coordinate is independent of h , because the motion normalizes V . The factor $d_i(q)$ is a left translation in V . It remains in a fixed compact set on the comparisons in question: both plaque-coordinate ranges and c are fixed. For the varying motion, right multiplication and its inverse have differential norm at most $\exp(O(R))$ in a left invariant metric on G and the induced quotient metric. Fixed chart changes have bounded differentials. Consequently both $\theta_i(q)$ and $d_i(q)$ vary over an input cell by

$$O(2^{-F} \exp(O(R))) = o(2^{-l_-}). \quad (4.8)$$

For reparametrized charts, the input domains $B(q)$ also vary by $O(2^{-F})$ over the cell, and their admitted output ranges vary by $O(2^{-F} \exp(O(R)))$. The fixed boundary margins absorb these variations. Thus the same estimate applies to those charts. The same bounds hold for pullbacks of paths in the extended charts. They justify following one plaque sheet throughout its prescribed range, rather than including a different sheet at a return of the orbit. They also justify the fitting assertion: vary q along the short path in its input cell and then follow the fixed plaque range inside the output extension.

On one such patch, θ_i is injective. To see this, the admitted output plaque ranges have a common open subset, by the margin and the small variation of $d_i(q)$. If two transverse inputs had the

same image, one could choose the same output plaque point in this common subset, contradicting injectivity of the motion and of the input chart. This observation permits product trims in output coordinates, even if many distinct input patches return to the same output box.

Trimming to fine columns and fixed plaque bins. Discard image patches meeting a coarse transverse wall. By (4.8), every point of such a patch is in a collar whose relative width tends to zero. The grid convention, applied to the output distribution μ_i , makes the discarded mass $o(1)$. Each remaining patch lies transversely in one coarse cell E .

Take fine transverse cells whose closures lie in its projected transverse range, and plaque bins admitted over the whole range. Their products form an inner set as in Lemma 4.4. Here are the two loss estimates. First, a path varying the transverse output coordinate through one fine cell pulls back a distance

$$O(2^{-l_+} \exp(O(R))) = o(2^{-F}). \quad (4.9)$$

Away from the input cell walls and a fixed small collar of the input chart boundary, it therefore stays in the input transverse range and in the chart. Input grid-wall loss is $o(1)$ by the selected shifts; chart-boundary loss can be made arbitrarily small by its null limiting mass. This verifies that almost all sampled fine transverse cells are admitted.

Second, on each plaque the transformation in (4.7) has bounded distortion, uniformly in i , and its dependence on q tends to zero over the patch. A sample away from a small collar of the input plaque boundary thus lies in an output plaque bin admitted over the whole transverse range, provided the fixed bins are chosen sufficiently fine. The collar has arbitrarily small sampling mass, again by the fixed null-boundary choice. For a reparametrized chart these collars are pulled back to its original fixed coordinate boundaries, which have zero limiting mass. Continuous tests expressed in the fixed coordinates pull back with uniform continuity moduli on the compact chart ranges. This gives arbitrarily small total trim loss. No bound on the number of patches is involved.

Comparing the conditional laws. The image of μ_i restricted to an input patch is exactly μ_i restricted to its image, by A invariance (or trivially for the identity motion). Apply Lemma 4.4 to its trim. The input patch law, converted by (4.7), agrees with P_E restricted to the admitted bins, up to arbitrarily small weighted error.

To express this without dividing by a possibly small admitted mass, take two continuous displacement tests f, ϕ supported with margin inside the input plaque range. Their output bin approximations are wholly inside the admitted bins. The unnormalized comparison says that each input test integral, multiplied by $P_E(B_p)$, agrees with the corresponding output P_E integral. Cross-multiplication therefore gives

$$P_{\text{in}}(f)P_{\text{out}}(\phi \circ c^{-1}) - P_{\text{in}}(\phi)P_{\text{out}}(f \circ c^{-1}) \longrightarrow 0 \quad (4.10)$$

in probability on the compared samples. All integrals here are in displacement coordinates centered at the respective samples. The small q variation changes the tests by a uniformly vanishing amount; sample-dependent centering is permitted by the uniform test assertion in Lemma 4.4. The input depth- F law can be replaced by its recorded probability by Lemma 4.2; the output coarse law can be replaced by the output recorded probability by the same lemma. This proves (4.10) for the data that actually define the states.

For completeness, the errors just used are localized as follows. Fix a pair of charts and open conditions placing the samples in inner boxes, requiring the full-plaque margin (4.6), and placing the displacement test supports inside the input plaque range with positive margins. The patch fits and trims hold for all sequence samples satisfying these conditions, uniformly for the chosen motion bound. Trim accuracy may be made arbitrarily good independently of the fixed cover or these

margin conditions. Thus the cross-product discrepancy tends to zero in probability on each such event. Exhaust by stricter margins and use continuous tests on compact inner boxes. Every joint distributional limit satisfies the corresponding exact identity there, since its tests and coordinate changes are continuous. Countably many charts, margin conditions, and tests suffice.

Gluing along one plaque. Apply this result first to the identity motion. Compare any chart at the sample with a chart extending far enough to contain its bounded plaque domain. By Lemma 4.3, each probability has positive mass on every neighborhood of the sampled point. Choose a nonnegative continuous test supported in a common such neighborhood and positive at that point. Its integral is positive for both probabilities. The cross-product identities then give projective agreement for every continuous test with compact support in the first plaque domain.

To compare two bounded domains, use a single extension containing both, on the same plaque sheet. Exhaust V by compact connected coordinate ranges containing 1 in their interiors. Successive extensions agree after normalization near 1, and so define a locally finite measure on the union. Local finiteness follows because each compact set lies in one extension carrying a finite chart probability with a finite positive normalization factor. The identity lies in its support by the sampling lemma. This construction does not require chaining through intermediate support points: each comparison uses a chart containing both the identity neighborhood and the entire displacement range in question. Using countably many tests and choosing normalizing bumps from a countable collection makes the resulting projective measure measurable.

Equivariance, including the long motion. For a fixed diagonal motion, the same cross-product identities give pushforward by $c(v) = a(-b)va(b)$, first on bounded test ranges and then globally by the extensions just constructed. For the varying motion in $\ker \beta$, c is the identity, and all estimates above are uniform for $\|b_i\| \leq CR_i$: the only growing chart cost is $\exp(O(R_i))$, dominated by (4.5). In a paired-state limit the second base point need not be a fixed function of the first. This causes no difficulty. The local identities hold on every paired open margin event, and these events exhaust probability one because both marginals admit the required extended charts. Their exact identities therefore hold in every such limit. Since the argument applies to an arbitrary sequence b_i under the displayed bound, it proves the last assertion with the stated quantifiers. $\square$

The measures m_ω^V retain information obtained by conditioning the original sequence, while having the chart consistency and diagonal comparison properties of ordinary plaque measures. The next step is to relate the data for different pattern groups; that requires a product argument, rather than a further change of coordinates.

5 A product rule for the plaque probabilities

The projective measures constructed in the preceding section retain information obtained by conditioning the measures μ_i at depths tending to infinity. We now prove a product rule for these measures. The factors in the rule belong to the *same augmented state*; this point will allow us to compose root directions in the next section.

Product decompositions for genuine leafwise conditional measures were developed by Einsiedler and Katok [4, Proposition 8.3] [5, Theorems 8.4–8.5]; see also Einsiedler and Lindenstrauss [6, Corollary 8.8 and Theorem 9.8]. The use of a diagonal element that contracts one factor and fixes another follows this established method. Here the measures are the augmented-state data obtained by conditioning before weak convergence. We prove the product identity for those data using the preceding mesh comparisons; the cited leafwise theorems are not applied to them.

We continue to use the notation $p \prec q$ for $p_i/q_i \rightarrow 0$. Every scale used below lies in the plateau $[l_{-,i}, l_{+,i}]$, with room for the displayed separations and for additional separations at both ends. Expectations in a chart mean integration against the restriction of μ_i to that chart; thus all error estimates are weighted by the mass of the chart. A chart of zero limiting mass can be omitted.

Theorem 5.1 (Product rule). *Let V and W be pattern groups, let $U = U_\alpha$ be a one-dimensional root group, and suppose that multiplication gives a coordinate decomposition*

$$V = WU, \quad W \triangleleft V.$$

Suppose also that some $a_0 \in A$ fixes U by conjugation and that the automorphism

$$\Delta(w) = a_0^{-1} w a_0$$

strictly contracts W . Then, for almost every augmented state ω ,

$$m_\omega^V \propto ((w, u) \mapsto wu)_*(m_\omega^W \otimes m_\omega^U). \quad (5.1)$$

Here $\propto$ denotes equality up to multiplication by a positive constant.

The case $W = \{1\}$ follows from the plaque comparisons. Henceforth $W \neq \{1\}$. The root constraints defining a contracting logarithm are rational linear equalities and strict inequalities. We can therefore choose a_0 , and rescale its logarithm, so that in the matrix-entry coordinates of W ,

$$\Delta(w)_{jk} = 2^{-d_{jk}} w_{jk}, \quad d_{jk} \in \mathbb{N}, \quad d_{jk} > 0. \quad (5.2)$$

Whenever jk and $k\ell$ compose, their weights add: $d_{j\ell} = d_{jk} + d_{k\ell}$. This is also the compatibility that makes Δ an automorphism.

We use the simultaneous charts in the stock of the preceding section:

$$(q, w, u) \mapsto \psi(q)wu \quad (5.3)$$

on product domains. The transverse coordinates are q for V , (q, w) for U , and (q, u) for W . The last assertion uses normality of W ; its plaque parametrization may be a smooth reparametrization of the group coordinate. Compatible transverse grids are used when the same q occurs in two conditionings.

The proof has two parts. Contraction of W shows that the U coordinate has the same law in the V and U data, independently of the W coordinate. Identifying the remaining factor requires a different argument. A depth- S U -cell label contains only $O(S)$ information. When $S \prec T$, revealing this label can appreciably change the law of a fixed-length further W -tile refinement, conditional on its parent tile, at only a vanishing proportion of T successive stages. Expanding a selected parent tile then turns this refinement comparison into a comparison of bounded W tests.

5.1 The factor fixed by the contraction

Lemma 5.2. *In each chart (5.3), the limiting V -plaque probability is a product in the coordinates w, u . Its u marginal agrees with the U -plaque probability at the sampled state. Consequently there is a projective locally finite measure ν_ω^W on W such that*

$$m_\omega^V \propto ((w, u) \mapsto wu)_*(\nu_\omega^W \otimes m_\omega^U). \quad (5.4)$$

Proof. Choose interior scales $l_- \prec T \prec F \prec l_+$, and apply the diagonal motion $a_0^{[T]}$. In one fixed input chart, a patch specifies a q -cell at depth F , with the whole w and u ranges. A subpatch also specifies a w -cell at depth F . Denote these two labels by Q_F and W_F .

Fix an arbitrarily small permitted coverage loss. Take finitely many output U -charts, with inner boxes and sufficiently large plaque extensions, covering all but that loss of output probability. The law of the output point is still μ_i . An input patch whose image meets one of these inner boxes lies in the corresponding extended chart, for all sufficiently large i : its U range has bounded fixed size, its W displacements shrink by $O(e^{-cT})$, and its q variation contributes at most

$$O(2^{-F} e^{CT})$$

for fixed constants $c, C > 0$. Assign such patches to output charts and omit the others. The omitted mass is bounded by the coverage loss.

In an assigned U -chart, the transverse image of a patch has diameter

$$O(e^{-cT} + 2^{-F} e^{CT}) = o(2^{-l_-}).$$

The transverse projection is independent of u . Its restriction to the input (q, w) range is injective: the chart has a common open U range for all these inputs, and two equal transverse coordinates would otherwise give overlapping images of two different points of the injective input chart. In the output plaque coordinate, the input u interval is translated by a parameter varying by $o(1)$ across the patch. Thus patches and their subpatches admit a common collection of inner U bins.

Here are the trimming details needed for Lemma 4.4. A patch meeting a coarse output wall has its sampled point in a collar of relative width $o(1)$; the grid convention makes the total mass of these patches $o(1)$. Fine output transverse cells can be pulled back at cost at most e^{CT} . Since $l_+ \succ F$, the pullback variation is negligible compared with the input q - and w -meshes. Away from their grid walls, the whole fine cell therefore stays inside the prescribed input labels. Fixed small collars of the input chart boundary have arbitrarily small mass. After removing these collars, sufficiently fine fixed output U bins lie wholly in the admitted plaque interval, except for arbitrarily small sampling mass. One may first trim each subpatch into products of fine transverse cells and these common plaque bins, and take their union as a trim of the patch. Both levels of the comparison use the same coarse output cell and the same plaque bins.

The patch/subpatch conclusion of Lemma 4.4 now gives, for every continuous function f on the closure of the input u interval,

$$\int |\mathbb{E}_{\mu_i}(f(u) \mid Q_F, W_F) - \mathbb{E}_{\mu_i}(f(u) \mid Q_F)| \, d\mu_i \longrightarrow 0. \quad (5.5)$$

To obtain the limit zero, first make the bin and chart-collar losses arbitrarily small for the fixed finite cover, and then make the coverage loss arbitrarily small. The common translation of the u coordinate causes no extra error, because its variation tends to zero and f is uniformly continuous.

The plateau comparison replaces the displayed depths by the depths defining the state data. It follows that the u marginal of the V data is the U probability at the sample. For independence, let g be a bounded continuous function of w , with modulus of continuity ω_g . Approximate it by a constant function g_F on each W_F -cell. Conditional expectation, followed by the triangle inequality, gives

$$\begin{aligned} & \int |\mathbb{E}_{\mu_i}(f(u)g(w) \mid Q_F) - \mathbb{E}_{\mu_i}(f(u) \mid Q_F)\mathbb{E}_{\mu_i}(g(w) \mid Q_F)| \, d\mu_i \\ & \leq \|g\|_\infty \int |\mathbb{E}_{\mu_i}(f(u) \mid Q_F, W_F) - \mathbb{E}_{\mu_i}(f(u) \mid Q_F)| \, d\mu_i + 2\|f\|_\infty \omega_g(C2^{-F}). \end{aligned}$$

The right side tends to zero by (5.5). Passing to the state limit therefore gives

$$P^V(f(u)g(w)) = P^V(f(u))P^V(g(w))$$

where P^V denotes the limiting V -chart probability. A countable dense family of continuous tests proves that the limiting probability is a product.

For a sampled plaque point with coordinates w_s, u_s , the group displacement from that point to wu has W, U coordinates

$$\left(\text{Ad}_{u_s^{-1}}(w_s^{-1}w), u_s^{-1}u \right). \quad (5.6)$$

This changes the two coordinates separately, so the product assertion holds in displacement coordinates as well. Proposition 4.5 identifies overlapping chart probabilities projectively. Apply this comparison on product ranges containing the identity and exhausting $W \times U$; charts with the required plaque extensions are available. The resulting W marginals agree projectively and define ν_ω^W . This proves (5.4). $\square$

It remains to prove that $\nu_\omega^W = m_\omega^W$ projectively. The contraction just used does not shrink the U direction, so interchanging the names of the two factors would not prove this. We instead compare small W pieces before a reverse diagonal motion with bounded W ranges after that motion.

5.2 Nested tiles and their information cost

We first construct subdivisions of W compatible with Δ . Let Γ_W be the lattice of pattern matrices with integral entries. Haar measure on W is Lebesgue measure in its matrix entries, with Γ_W of covolume one.

Lemma 5.3 (Tiles adapted to the contraction). *There is a compact set $\mathcal{F} \subset W$, of Haar measure one and with Haar-null boundary, such that the translates $\gamma\mathcal{F}$, $\gamma \in \Gamma_W$, tile W with disjoint interiors. Moreover,*

$$\mathcal{F} = \bigcup_{d \in \mathcal{D}} \Delta(d)\Delta(\mathcal{F}), \quad \mathcal{D} = \{\gamma \in \Gamma_W : 0 \leq \gamma_{jk} < 2^{d_{jk}} \text{ for each pattern entry } jk\},$$

where d_{jk} is the contraction weight in (5.2). The union is disjoint off boundaries.

For every integer $k \geq 0$, the tiles

$$\gamma\Delta^k\mathcal{F}, \quad \gamma \in \Delta^k\Gamma_W,$$

form a subdivision, and these subdivisions are nested as k increases. They may all be shifted by the same left translation g . If g is distributed according to Haar probability on W/Γ_W , then, for every fixed $w \in W$, its coordinate in its shifted level- k tile has uniform Haar distribution on $\mathcal{F}$, after rescaling by Δ^{-k} . In particular, the probability of lying in a fixed ρ -collar of $\partial\mathcal{F}$ in these coordinates tends to zero as $\rho \downarrow 0$, uniformly in k and w .

Proof. For this proof write d_α for the positive integral weight of a pattern entry α , and put $B_\alpha = 2^{d_\alpha}$. The digit set consists of integral pattern matrices d whose α entry belongs to $\{0, \dots, B_\alpha - 1\}$. Define

$$\mathcal{F} = \left\{ \prod_{\nu=1}^{\infty} \Delta^\nu(d_\nu) : d_\nu \in \mathcal{D} \right\}, \quad (5.7)$$

with factors ordered by increasing ν . Contraction and the finite nilpotent multiplication formulas imply uniform convergence of these products. More explicitly, in any entry the product is a finite

sum of convergent series of products of digit entries, with positive geometrically decaying weights. Thus $\mathcal{F}$ is compact.

Order the pattern entries by increasing root height after putting the pattern in upper triangular form. In an entry α , the coordinate in (5.7) is

$$\sum_{\nu \geq 1} B_{\alpha}^{-\nu}(d_{\nu})_{\alpha} + C_{\alpha},$$

where C_{α} is determined entirely by digits in entries of smaller height. At height one, this is the usual base- B_{α} expansion of a number in $[0, 1]$. Inductively, once the lower-height digits have been specified outside their null sets of ambiguous expansions, the possible α coordinates form the interval $[C_{\alpha}, C_{\alpha} + 1]$, with a unique digit expansion except at its countably many base endpoints.

Left multiplication by an integral pattern matrix has the same triangular form: the α coordinate changes by its integral α entry plus a function of already chosen lower-height entries. Reduce entries in this order. At each step exactly one integer places the remaining coordinate in an interval of length one, except at an endpoint, and its base expansion is then unique almost everywhere. Fubini's theorem at each of the finitely many entries proves that almost every point of W belongs to exactly one translate $\gamma\mathcal{F}$. It also proves that $\mathcal{F}$ has Haar measure one.

The translates are locally finite, because $\mathcal{F}$ is compact and Γ_W is discrete. Their union is consequently closed, and the almost-everywhere covering just proved implies that they cover all of W . Splitting the first digit from (5.7) gives

$$\mathcal{F} = \bigcup_{d \in \mathcal{D}} \Delta(d)\Delta(\mathcal{F}). \quad (5.8)$$

Triangular reduction also shows that $\mathcal{D}$ represents the right cosets of $\Delta^{-1}\Gamma_W$ in Γ_W :

$$\Gamma_W = \bigsqcup_{d \in \mathcal{D}} (\Delta^{-1}\Gamma_W)d.$$

Applying Δ shows that the origins in the subdivision of all level-zero tiles are exactly

$$\Delta\Gamma_W = \bigsqcup_{d \in \mathcal{D}} \Gamma_W\Delta(d).$$

Iteration proves the asserted level- k subdivision and nesting.

For completeness, these compact tiles have the regularity needed for boundary trimming. The closed translates cover W , so the Baire category theorem gives nonempty interior to $\mathcal{F}$. Every point of $\mathcal{F}$ lies in a descendant tile in the iterated subdivision (5.8). Such descendants have nonempty interior and diameters tending to zero. Therefore $\mathcal{F}$ is the closure of its interior. An interior intersection of two translates would then give a positive-volume overlap, contradicting almost-everywhere uniqueness. A boundary point of one translate belongs to another: otherwise local finiteness would give a neighborhood covered only by the first translate, making that point interior. Thus the boundary is contained in the union of the pairwise overlaps, which is null. Boundary assignments can be chosen measurably and consistently through the nested subdivisions.

It remains to check the law of a shifted tile coordinate; this check matters because the sampling measure on W need not have a density. Since the weights are positive integers, $\Gamma_W \subset \Delta^k\Gamma_W$. The projection

$$W/\Gamma_W \longrightarrow W/\Delta^k\Gamma_W$$

sends Haar probability to Haar probability. Inversion and right multiplication by w send the latter distribution to invariant volume on the left quotient $\Delta^k\Gamma_W \backslash W$. The representative of $g^{-1}w$ in

$\Delta^k \mathcal{F}$ therefore has normalized Haar distribution on that tile, and rescaling gives Haar measure on $\mathcal{F}$. This is precisely the coordinate of w in its tile $g\gamma\Delta^k \mathcal{F}$. The null boundary implies the collar assertion by continuity from above. Fubini's theorem permits the same conclusion after sampling w from any probability measure. $\square$

For finite labels we write H for Shannon entropy, with natural logarithms, and

$$I(Y; Z \mid X) = H(Y \mid X) - H(Y \mid X, Z)$$

for conditional mutual information. Tile labels restricted to a fixed chart are finite at every fixed level.

Lemma 5.4 (Information in a transverse label). *Fix a chart (5.3) and integers $F, S \geq 1$. Let Q_F be its depth- F q -cell label, and let U_S be its depth- S u -cell label. Let K_k be the level- k tile label of w , with any common shift as in Lemma 5.3, restricted to the chart. For every pair of integers $T \geq j \geq 1$,*

$$\sum_{k=0}^{T-j} I_{\mu_i}(K_{k+j}; U_S \mid Q_F, K_k) \leq Cj(1+S). \quad (5.9)$$

Here the measure in the chart is normalized for defining entropy; its mass is restored when errors are integrated over the chart. The constant C depends only on the fixed coordinate domain, and is independent of i, F, S, T, j , the sampling measure, and the tile shift. Consequently, along any sequence with $T \rightarrow \infty$ and $S/T \rightarrow 0$, for every fixed j , averaged over $T/2 \leq k \leq 3T/4$, revealing U_S changes the conditional law of K_{k+j} , given (Q_F, K_k) , by $o(1)$ in expected ℓ^1 distance. The same assertions hold after averaging over shifts.

Proof. The number of u -cells meeting its fixed bounded interval is $O(2^S)$, so $H(U_S) \leq C(1+S)$. Because the tile labels are nested, the chain rule gives

$$\sum_{k=0}^{L-1} I(K_{k+1}; U_S \mid Q_F, K_k) = I(K_L; U_S \mid Q_F, K_0) \leq H(U_S).$$

For a jump of length j , expand its mutual information into these one-stage increments. Each increment occurs at most j times in the sum over starting stages, proving (5.9). Pinsker's inequality and Cauchy–Schwarz bound the average conditional ℓ^1 error. For $T \geq 4j$, write $\mathcal{K}_T = \{k \in \mathbb{Z} : T/2 \leq k \leq 3T/4\}$. The bound is

$$\begin{aligned} & \frac{1}{|\mathcal{K}_T|} \sum_{k \in \mathcal{K}_T} \mathbb{E} \|\mathcal{L}(K_{k+j} \mid Q_F, K_k, U_S) - \mathcal{L}(K_{k+j} \mid Q_F, K_k)\|_1 \\ & \leq C \sqrt{\frac{j(1+S)}{T}}. \end{aligned}$$

This tends to zero for fixed j . The entropy bound is uniform in the shift, so it can also be averaged over shifts. $\square$

5.3 Identification of the contracted factor

We now compare ν_ω^W with m_ω^W . Choose interior scales

$$l_- \prec S \prec T \prec F \prec l_+. \quad (5.10)$$

Fix nonnegative functions $f, \phi \in C_c(W)$, bounded by one, with $\phi(1) > 0$. It suffices to show

$$\frac{\nu_\omega^W(f)}{\nu_\omega^W(\phi)} = \frac{m_\omega^W(f)}{m_\omega^W(\phi)} \quad (5.11)$$

almost surely. Identity support makes the denominators positive. A countable determining family of such tests will then give projective equality.

The comparison is made at the output of the motion

$$a_0^{-(k+m_0)}, \quad (5.12)$$

where $T/2 \leq k \leq 3T/4$ and m_0 is a fixed positive integer. A level- k input tile has, relative to its translated origin, the fixed output shape $\Delta^{-m_0}\mathcal{F}$. The role of m_0 is to make this shape contain the supports of the tests about most sampled points. The role of the much finer level $k+j$, with j fixed after m_0 , is to resolve those tests at output.

The two distributions to be compared can now be specified before the geometric details. Inside one input chart, a parent patch has label $p = (Q_F, K_k)$, leaving the whole u interval available; a subpatch has label $p' = (Q_F, K_k, U_S)$. Lemma 5.4 compares

$$\mathcal{L}_{\mu_i}(K_{k+j} \mid Q_F, K_k) \quad \text{and} \quad \mathcal{L}_{\mu_i}(K_{k+j} \mid Q_F, K_k, U_S). \quad (5.13)$$

The measure here is restricted to the input chart. After (5.12), the parent W tile has fixed shape $\Delta^{-m_0}\mathcal{F}$, and its descendants have shape $\Delta^{j-m_0}\mathcal{F}$. Increasing the fixed j therefore allows (5.13) to compare bounded continuous W tests on that output range.

In a simultaneous output chart with coordinates (q', w', u') , the transverse coordinate of the whole patch is q' ; for its thin u -subpatch it is (q', u') . The geometry and plateau estimate below will consequently compare the whole-patch test law with a restriction of the V data, and the subpatch test law with a restriction of the W data. These comparisons have normalization factors and retain an inner u' -range in the V data. We will first compare their cross products, then use Lemma 5.2 to remove that u' restriction, and finally control the denominators. This is how the two conditional tile laws will identify the W factor without presupposing the desired projective equality.

Figure 1 contrasts these two comparisons: contraction determines the U factor, whereas expansion must be combined with the information estimate to identify the W factor.

Coverage and the order of choices. Fix a permitted exceptional probability $\epsilon > 0$. Take finitely many input charts (5.3), with inner boxes and fixed margins, covering all but a sufficiently small multiple of ϵ . Estimates can be added over these charts; their finite multiplicity only changes constants. Use the patches and subpatches just described, with whole level- k w -tiles. Tiles meeting the fixed inner boxes lie inside the outer input w ranges for large i , because their diameters tend to zero.

First choose a tile-coordinate collar thin enough that its expected mass is much smaller than the permitted coverage loss. Lemma 5.3 does this uniformly in k and in the input distribution. Outside this collar the sample has a fixed positive margin in $\mathcal{F}$. Choose m_0 so large that every displacement in a fixed compact set containing $\text{supp } f \cup \text{supp } \phi$ and the identity lies, with margin, in the output tile about such a sample. Indeed, pull such a displacement back to the input tile coordinate. It is contracted by Δ^{m_0} ; the conjugation coming from the input u coordinate is uniformly bounded, and commutes with Δ , since a_0 fixes U . Thus the pulled-back displacement fits inside the chosen margin for sufficiently large fixed m_0 . The input inner u intervals also give a fixed U neighborhood about the sampled output point.

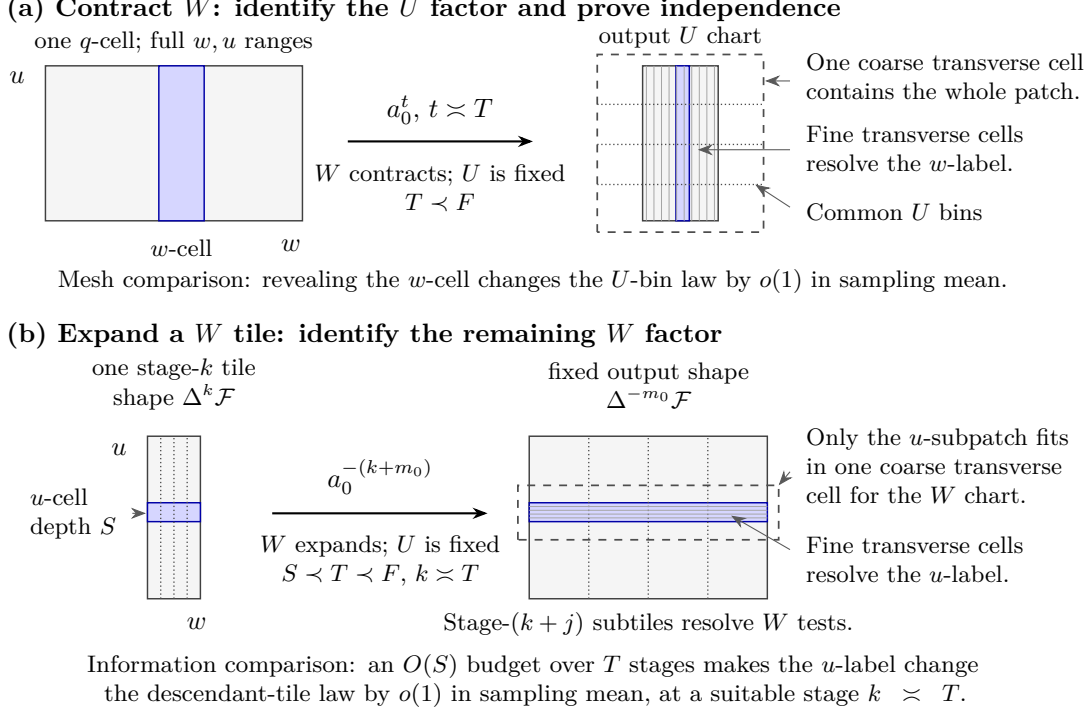


Figure 1: The two different comparisons in the product rule. Each input picture fixes a q -cell at depth F ; the coordinate q is suppressed. Dashed outlines denote coarse output transverse cells; thin solid lines denote fine transverse cells. Dotted lines denote fixed U bins in (a) and stage- $(k+j)$ W subtiles in (b). Shading marks a labelled subpatch, not a density. In (a), contraction permits one coarse transverse-cell comparison for the whole patch and its subpatches. In (b), expansion leaves the full U interval broad. The full patch is compared with V data and the thin u -subpatch with W data; the information estimate compares their descendant-tile laws and hence their resolved W tests. All widths and tile shapes are schematic: W can have several dimensions and its tiles need not be rectangles. The resulting factor identifications are projective.

Now take finitely many output charts of the same simultaneous form, with inner boxes and V -extensions large enough for the images of these patches. Such a finite cover loses an arbitrarily small further probability. For fixed m_0 , all displacements from a sample to points in its output patch lie in a fixed compact range: use the tile origin as base in W , a fixed base in U , and the fixed shape $\Delta^{-m_0}\mathcal{F}$. The q variation after (5.12) is $O(2^{-F}e^{CT})$. Hence a patch meeting an output inner box fits in its extended chart with margin. We can assign these patches to the finitely many output charts, or perform the estimates for each applicable input–output pair. We keep only samples with the indicated inner margins.

These choices fix the chart cover, the output tile extent, and the test-range margins. They do *not* yet fix the accuracy of bin comparisons. Afterwards one may make a second, arbitrarily thinner tile-boundary exclusion, refine the fixed output bins, and increase the fixed integer j . These later choices improve errors without changing the earlier coverage requirements.

All these requirements can be met simultaneously with the information estimate. Initially average both over shifts and over $k \in [T/2, 3T/4]$. The first collar was chosen to have expected loss small enough that its required coverage bound holds with high probability. Any later, finer collar can have arbitrarily small expected loss. For every fixed j , Lemma 5.4 gives an error tending to zero in this joint average. Markov’s inequality therefore selects common shifts and $k = k_i$ satisfying the coverage bounds and the desired comparison-error bounds. The transverse grids in q, u may be fixed using the grid convention of the preceding section. Every scale still satisfies (5.10).

Geometry of the output comparisons. Write the coordinates of an assigned output chart as (q', w', u') . For fixed input q , its patch image is obtained by left multiplication in V from

$$\Delta^{-m_0}\mathcal{F} \times (\text{input } u \text{ interval}).$$

Left multiplication by w_0u_0 sends wu to

$$(w_0 \text{Ad}_{u_0}(w))(u_0u). \quad (5.14)$$

Thus the two plaque coordinates change separately. The parameters of this bounded left multiplication vary by $O(2^{-F}e^{CT})$ across the input q -cell. The V -transverse diameter in q' is bounded by the same quantity. It is smaller than 2^{-CS} for every fixed C when i is large. For a subpatch, the W -transverse coordinates are (q', u') ; their diameter is

$$O(2^{-S} + 2^{-F}e^{CT}) = o(2^{-l-}).$$

Discard patches or subpatch comparisons crossing a coarse output transverse wall. Their sampling mass is $o(1)$.

We apply Lemma 4.4 to V data on the patches and W data on the subpatches. The needed inner products are as follows. For a patch take collections of w' and u' bins in product form; for each of its subpatches use the same inner collection of w' bins. These collections are admitted over the entire corresponding transverse projection, by (5.14) and the vanishing variation of its parameters.

We verify that these inner products lose as little sampling mass as desired. The extended charts describe one plaque sheet throughout each of the ranges involved. The transverse maps are injective on a patch or subpatch: their images have a common open plaque range, and injectivity follows from injectivity of the original chart and the diagonal motion. Vary a fine output transverse coordinate across its cell and pull the path back. Its input length is at most e^{CT} times the fine mesh side. Because $F, S \prec l_+$, this path stays in its input q - and u -cells except at grid-wall collars of mass $o(1)$. Fixed input chart-boundary collars have arbitrarily small mass. In tile coordinates the same

pullback costs at most $e^{C'T}$, so it stays in the input tile away from any fixed thin collar of that tile's boundary. The second collar chosen above makes this loss arbitrarily small.

Finally, away from these collars, the sample has a definite margin in its admitted w' and u' ranges. For w' the margin depends on m_0 , the fixed charts, and the tile collar thickness; for u' it follows from the fixed interval margin. Sufficiently fine fixed plaque bins are then wholly admitted. This proves the required plaque-bin trim estimate. The collections can retain the supports of f, ϕ about the covered samples and a fixed U neighborhood about their u' coordinates. Thus the comparison errors can be made arbitrarily small after the original coverage and range choices.

Comparison of the W tests. At a sampled output point, let P_V and P_W denote its finite- i chart probabilities for V and W . Interpret f, ϕ in the output w' coordinate using the displacement formula (5.6). Their parameters may depend on the sample, but their norms and continuity moduli are uniform on the finitely many chart ranges. Let B_u denote the union of the inner u' bins for its patch. We claim that, on the covered samples,

$$P_V(f 1_{B_u})P_W(\phi) - P_V(\phi 1_{B_u})P_W(f) \quad (5.15)$$

has arbitrarily small expectation of its absolute value in the limit $i \rightarrow \infty$.

Here is the mass calculation. Let A_p be the law of the output plaque coordinates when sampling the input patch p , and let $A_{p'}$ be the corresponding law for its subpatch. The unnormalized form of Lemma 4.4, first for V and then for W , gives

$$\begin{aligned} c_p A_p(f) &\simeq P_V(f 1_{B_u}), & c_p A_p(\phi) &\simeq P_V(\phi 1_{B_u}), \\ c_{p'} A_{p'}(f) &\simeq P_W(f), & c_{p'} A_{p'}(\phi) &\simeq P_W(\phi). \end{aligned}$$

Let $e_f(x), e_\phi(x), e'_f(x), e'_\phi(x)$ denote the absolute errors in these four comparisons at the sampled point x . The factors $c_p, c_{p'}$ are the probabilities of the admitted plaque-bin ranges in the respective coarse output laws, hence belong to $[0, 1]$. Each $\simeq$ means an absolute error whose sum weighted by the relevant patch masses is arbitrarily small. The tests are supported in the common inner w' range, so there is no additional w' -restriction in the right-hand integrals. Replacing coarse laws by the data laws uses the plateau comparison. These estimates remain true for the sample-dependent test parameters by the uniform-test clause of Lemma 4.4.

At input, Lemma 5.4 compares the law of the level- $(k+j)$ tile in p with its law in p' . At output these finer tiles have relative shapes $\Delta^{j-m_0}\mathcal{F}$. Once m_0 and the finite chart ranges are fixed, taking the fixed integer j large makes their diameters as small as prescribed. The conversion to w' coordinates has uniformly bounded distortion and is independent of u , by (5.14). Let $\eta_i(x)$ be the ℓ^1 distance between the two conditional level- $(k+j)$ tile laws for the patch and the subpatch containing x . There is a uniform bound

$$\sigma_{i,j} \leq C \operatorname{diam}(\Delta^{j-m_0}\mathcal{F}) + O(2^{-F}e^{CT})$$

for the variation of w' within one such tile, including the variation across its q -cell. If h is either f or ϕ , and ω_h is a common continuity modulus for its chart-coordinate versions, then

$$|A_p(h) - A_{p'}(h)| \leq \eta_i(x) + 2\omega_h(\sigma_{i,j}). \quad (5.16)$$

Indeed, replace h by one value on each finer tile. The two replacements cost at most $\omega_h(\sigma_{i,j})$ each, and the remaining finite-label difference is bounded by $\eta_i(x)$. This also proves the assertion for tests based at the sampled point, since the bound is uniform over the test coefficients. The expected value of η_i tends to zero for the selected shifts and stages, by Lemma 5.4.

Multiply the four mass comparisons and cancel by cross multiplication, without dividing by c_p or $c_{p'}$. The result is (5.15): the remaining expression is bounded by

$$e_f(x) + e_\phi(x) + e'_f(x) + e'_\phi(x) \\ + 2\eta_i(x) + 2\omega_f(\sigma_{i,j}) + 2\omega_\phi(\sigma_{i,j}).$$

For this inequality, the term left after the four mass comparisons is $c_p c_{p'} (A_p(f)A_{p'}(\phi) - A_p(\phi)A_{p'}(f))$; apply (5.16) and use that all factors are bounded by one. Integrating over the covered samples bounds the first four terms by the total trim and bin-comparison errors, the fifth by the stage-information error, and the last two by their uniform moduli. Choose the trim accuracy and fixed bins first, then choose the fixed j sufficiently large, and finally let i tend to infinity. This proves the claimed integrated bound for (5.15). The number of patches may grow with i ; the absolute mass estimate (4.3) charges each fine-cell/bin pair at most once within a fixed chart comparison.

Removing the U restriction. By Lemma 5.2, in the limiting V data the w' and u' coordinates are independent. Consequently, for the finite- i probabilities,

$$P_V(f1_{B_u}) - P_V(f)P_V(B_u) \longrightarrow 0$$

in probability on the charts under consideration, and the same holds with ϕ . To justify the varying B_u , its bins come from one fixed finite subdivision, and every bin boundary has zero mass almost surely in the limiting data. The asserted convergence therefore holds simultaneously for all subsets of that finite bin collection. The displacement tests depend continuously on the sample on the fixed inner chart ranges. Output sampling has law μ_i , so the limiting independence applies there even though the patches were chosen at input. If several input charts are used, their finite multiplicity only multiplies the error bound.

Thus (5.15) becomes

$$P_V(B_u)(P_V(f)P_W(\phi) - P_V(\phi)P_W(f)) \simeq 0. \quad (5.17)$$

Denominators and passage to the projective limit. We now explain why (5.17) gives (5.11). On each of the fixed covered chart ranges, B_u contains a fixed U neighborhood about the sample. Choose a continuous bump supported in that neighborhood and positive at the sampled u' coordinate. Its integral in the limiting V probability is positive by identity support. The same support property gives positivity of the limiting integrals $P_V(\phi)$ and $P_W(\phi)$. For any further prescribed exceptional probability there is, therefore, a constant $\kappa > 0$ such that, for all large i ,

$$P_V(B_u) \geq \kappa, \quad P_V(\phi) \geq \kappa, \quad P_W(\phi) \geq \kappa \quad (5.18)$$

outside that exception on the covered samples. This follows from convergence of the joint data in the finitely many charts, using the fixed continuous bumps.

The order of choices is essential. Fix the coverage, m_0 , the output charts, and their test-range margins first. Then choose κ from these fixed bumps. Only afterwards refine the bin approximations and tile collars used for comparison, and enlarge j , to make the error in (5.17) as small as required relative to κ^3 . The coverage collar and the fixed neighborhoods giving the lower bounds need not change. Division in (5.17) is now legitimate and yields arbitrarily small difference between

$$\frac{P_V(f)}{P_V(\phi)} \quad \text{and} \quad \frac{P_W(f)}{P_W(\phi)}$$

outside an arbitrarily small exceptional probability. Explicitly, let $\mathcal{G}_i$ be the covered set on which (5.18) holds, and denote the expression on the left of (5.17) by $D_i(x)$. For every $\delta > 0$,

$$\mu_i \left(\mathcal{G}_i \cap \left\{ \left| \frac{P_V(f)}{P_V(\phi)} - \frac{P_W(f)}{P_W(\phi)} \right| > \delta \right\} \right) \leq \frac{1}{\delta \kappa^3} \int_{\mathcal{G}_i} |D_i(x)| d\mu_i(x).$$

The right side can be made arbitrarily small with the previously fixed coverage and denominators.

To see that this proves the assertion for the limiting projective measures, suppose otherwise. A countable choice of tests and charts would then give a reference output chart and a positive-probability set on which the two ratios differ by more than some fixed positive number. Restrict this set so that the denominators have a positive lower bound and the supports of the tests, together with a U neighborhood, have inner margins in the chart. Continuous displacement integrals and the open-set inequality preserve a positive probability of discrepancy in the sequence data.

Apply the preceding coverage construction with exceptional probability much smaller than this discrepancy probability. On samples covered by both the reference chart and an assigned output chart, Proposition 4.5 and Lemma 5.2 identify their limiting ratios. This also identifies their finite- i ratios up to an error tending to zero outside a further small set: first impose closed inner margins and lower bounds on the denominators, then use joint convergence of the finitely many chart data. The arbitrarily accurate comparison just proved in the assigned charts contradicts persistence of the discrepancy in the reference chart.

We have proved (5.11). Taking countably many tests and then exhausting compact W ranges gives $\nu_\omega^W \propto m_\omega^W$ almost surely. Together with (5.4), this completes the proof of Theorem 5.1.

6 Active roots and their support

The product rule turns the plaque measures into a directed graph at each augmented state. We first show that its arrows compose and describe the support of the positive-root plaques. We then realize this graph by measurable indicators on the original spaces, retaining the long-wall comparison needed for the entropy argument.

Write $\mathbf{P}$ for the law of the limiting augmented state $\omega = (x, (P^B)_B)$. For $j \neq k$, define

$$a_{jk}(\omega) = \mathbf{1}_{\{m_\omega^{U_{jk}} \text{ is not supported on } \{1\}\}}.$$

We call jk *active* when $a_{jk} = 1$. This definition does not depend on the representative of the projective measure. It is measurable: after normalizing by a positive integral against a bump near the identity, activity is detected by a countable family of nonnegative continuous tests supported away from the identity. All assertions below are made on a common set of full $\mathbf{P}$ -measure on which the comparison and product rules hold.

Proposition 6.1 (Composition and positive-root support). *Almost surely, activity is transitively closed:*

$$a_{pq}(\omega)a_{qr}(\omega)(1 - a_{pr}(\omega)) = 0 \quad (p, q, r \text{ distinct}).$$

Let $b \in L$ be regular, let V be the pattern group of all b -positive roots, and let V_ω^+ be the subgroup generated by the active b -positive root groups. These active roots form a pattern, and

$$\text{supp } m_\omega^V \subset V_\omega^+.$$

Both conclusions hold simultaneously for every regular b .

Proof. First observe that Theorem 5.1 factors the measure of any pattern group into its one-dimensional root measures, in a suitable order. Indeed, the cone generated by its roots is pointed. Choose a root α on an exposed extremal ray. No other root in the pattern is proportional to α , so a diagonal logarithm can vanish on α and be strictly negative on every remaining root. Removing α leaves a normal pattern subgroup W : a sum of two roots in the original pattern cannot equal α , since extremality would force both summands onto its ray. Thus the hypotheses of Theorem 5.1 hold for $V = WU_\alpha$. Iterate, allowing $W = \{1\}$ at the last step.

To prove transitivity, fix distinct p, q, r and consider the three-root Heisenberg group generated by U_{pq}, U_{qr}, U_{pr} . Suppose that $a_{pr} = 0$. Factoring first with U_{pq} as the final factor, and then with U_{qr} as the final factor, gives two descriptions of the same projective measure. Write $u_{jk}(t) = 1 + tE_{jk}$. Since the U_{pr} factor is a point mass, the first description is supported on the matrices

$$u_{qr}(t)u_{pq}(s) = 1 + sE_{pq} + tE_{qr},$$

whereas the other is the pushforward of the product of the two root measures under

$$u_{pq}(s)u_{qr}(t) = 1 + sE_{pq} + tE_{qr} + stE_{pr}.$$

If both roots were active, each root measure would give positive mass to some compact parameter set disjoint from zero. Their product would then give positive mass to matrices with nonzero pr entry, contrary to the first description. The projective identities apply on compact ranges containing these sets, so all masses used here are finite. This proves transitivity.

For the support assertion, use the successive factorization already proved. Every inactive root factor is a point mass at the identity; every other factor is supported on its own active root group. The product is therefore supported on the subgroup generated by the active roots. Transitivity says exactly that the active positive roots are closed under root addition, so this subgroup is a pattern group and is closed. There are only finitely many positive-root patterns, which also gives the simultaneous assertion. $\square$

The entropy argument will use plaques enlarged by the diagonal group. The following consequence retains the activity at the sampled state.

Lemma 6.2 (Support on plaques enlarged by A). *With V and V_ω^+ as in Proposition 6.1, the limiting VA -plaque probability, written in displacement coordinates from its sampled point, is supported on V_ω^+A within its chart.*

Proof. Use a product box with coordinates $\psi(q)va$, with product coordinate domains. Local A -invariance gives, for each i , a product decomposition

$$d\mu_i(q, v, a) = d\nu_i(q, v) da$$

inside the box, where da is Haar measure restricted to its coordinate box and the normalization is absorbed into ν_i . For VA plaques the transverse coordinate is q ; for V plaques it is (q, a) . Choose a common q grid and supplement it by an a grid for the latter conditioning. Conditional on a q cell, the VA probability is the product of a probability on v and normalized Haar measure on a . Conditioning also on an a cell leaves the v probability unchanged. These are exact identities before taking a limit. The interior-depth comparisons permit these compatible grids in the defining data, and the identities pass to the joint limit by continuous tests and the null-boundary convention.

At a sample $x = \psi(q_s)v_s a_s$, the V displacements corresponding to varying v at fixed a_s are

$$a_s^{-1}v_s^{-1}va_s.$$

By Proposition 6.1, their probability is supported on V_ω^+ . A general VA displacement is

$$a_s^{-1}v_s^{-1}va = (a_s^{-1}v_s^{-1}va_s)(a_s^{-1}a),$$

and hence belongs to V_ω^+A almost surely. The argument applies on inner chart boxes; exhausting them gives the statement on the whole plaque box, whose boundary has zero probability. $\square$

Lemma 6.3 (Indicators along the sequence). *Fix an interior scale R_i allowed in the long-wall comparison of Proposition 4.5, with ratio-separated depths on both sides of it inside $(l_{-,i}, l_{+,i})$. There exist measurable functions $g_{jk,i} : X \rightarrow \{0, 1\}$ such that*

$$\text{Law}_{\mu_i}(Z_i(x), (g_{jk,i}(x))_{j \neq k}) \implies \text{Law}_{\mathbf{P}}(\omega, (a_{jk}(\omega))_{j \neq k}).$$

They satisfy asymptotic transitivity:

$$\int_X g_{pq,i}(x)g_{qr,i}(x)(1 - g_{pr,i}(x)) d\mu_i(x) = o(1) \quad (p, q, r \text{ distinct}),$$

and, for every fixed $C < \infty$,

$$\sup_{\substack{b \in \ker \beta_{jk} \\ \|b\| \leq CR_i}} \int_X |g_{jk,i}(xa(b)) - g_{jk,i}(x)| d\mu_i(x) \longrightarrow 0. \quad (6.1)$$

Proof. Let $\mathcal{S}$ be the separable metric state space used for Z_i , and put $A_{jk} = \{\omega : a_{jk}(\omega) = 1\}$. Choose $\epsilon_r \downarrow 0$. For each root and each r , there is a $\mathbf{P}$ -continuity set $D_{jk,r} \subset \mathcal{S}$ such that

$$\mathbf{P}(D_{jk,r} \triangle A_{jk}) < \epsilon_r.$$

For completeness, use a countable family of continuous coordinate tests generating the Borel sets of $\mathcal{S}$. For each test choose a countable dense family of thresholds outside the atoms of its distribution under $\mathbf{P}$. The finite Boolean combinations of the resulting threshold sets have null boundary and approximate every measurable set in $\mathbf{P}$ -measure. Thus the approximations may be made using finitely many data tests at each stage.

For fixed r , set $g_{jk,i}^{(r)}(x) = \mathbf{1}_{D_{jk,r}}(Z_i(x))$. Weak convergence of the states and the null boundaries give joint convergence of the states and these finitely many indicators. Since the exact indicators are transitive, the limiting transitivity defect for any triple is at most $3\epsilon_r$.

We next prove the uniform wall estimate at this fixed approximation level. Fix a root, C , and any sequence $b_i \in \ker \beta_{jk}$ with $\|b_i\| \leq CR_i$. Both marginals of

$$(Z_i(x), Z_i(xa(b_i)))$$

have the ordinary state law, because μ_i is A -invariant. Consequently these paired laws are tight. In every subsequential limit their two marginals are $\mathbf{P}$, and Proposition 4.5 identifies the two projective U_{jk} measures. Their exact activity indicators are therefore equal. Applying the continuity-set approximation separately in the two marginals shows that every such limit has

$$\mathbb{E}|\mathbf{1}_{D_{jk,r}}(\omega) - \mathbf{1}_{D_{jk,r}}(\omega')| \leq 2\epsilon_r.$$

The boundary of the tested event is contained in the union of the two marginal boundaries, so the corresponding prelimit expectations converge along this subsequence. Since the sequence b_i was arbitrary, a sequence chosen within a vanishing error of the supremum proves

$$\limsup_i \sup_{\substack{b \in \ker \beta_{jk} \\ \|b\| \leq CR_i}} \int_X |g_{jk,i}^{(r)}(xa(b)) - g_{jk,i}^{(r)}(x)| d\mu_i(x) \leq 2\epsilon_r.$$

This argument uses limits of paired states even when b_i diverges.

Finally choose $r = r(i) \rightarrow \infty$ sufficiently slowly. At stage r impose the preceding estimates for all roots, all triples, and all integers $1 \leq C \leq r$, together with the first r tests of a convergence-determining family for the joint state and indicator law. The limiting laws at fixed r differ from the exact activity law by at most the sum of the finitely many approximation errors. A diagonal choice therefore gives the asserted joint convergence, vanishing transitivity defects, and (6.1) for $g_{jk,i} = g_{jk,i}^{(r(i))}$. Bounds for integral C imply the bounds for every fixed finite C . $\square$

We conclude with the relation between activity and pieces having countable intersections with root plaques. It is this relation that will exclude proper homogeneous components once all roots have been shown to be active.

Lemma 6.4 (Inactivity on pieces with countable plaque sections). *Let $S \subset X$ be a Borel set whose intersection with each local U_{jk} plaque is countable. Then*

$$a_{jk}(\omega) = 0 \quad \text{for } P\text{-almost every state with } x \in S.$$

In particular, this applies to the fixed-point pieces of Lemma 3.4 for roots crossing distinct eigenvalues: each such piece meets a sufficiently small root plaque in at most one point. It also applies to countable unions of these pieces.

Proof. Work first in a U_{jk} flow box with coordinates (q, h) and plaque probability P^B . For each q , the section

$$S_q = \{h : \psi(q)h \in S\}$$

is countable. The sampling identity for this single chart states that the conditional law of the sampled h , given (q, P^B) , is P^B . A probability measure assigns mass zero to those points of a countable set which are not its atoms. Consequently

$$P\{x \in S \cap B, P^B(\{h\}) = 0\} = 0.$$

No conditioning on the other plaque probabilities is used here. Projective agreement in the chart identifies an atom at the sampled h with an atom at the identity of $m_\omega^{U_{jk}}$. A countable flow-box cover therefore shows that this root measure has an atom at the identity almost surely over S .

It remains to show that, under the full state law, an atom at the identity forces the entire root measure to be supported there. Identify U_{jk} with $\mathbb{R}$ by $t \mapsto 1 + tE_{jk}$. Choose a fixed $b \in L$ with $\lambda = e^{\beta_{jk}(b)} > 1$. Fixed-motion equivariance in Proposition 4.5, together with A -invariance of the prelimit sampling law, says that the projective law of $m_\omega^{U_{jk}}$ is invariant under the pushforward by $t \mapsto \lambda t$.

The event that this measure has an atom at zero is itself invariant under dilation. If that event has positive probability, condition the projective-measure law on it and normalize each measure m by its atom: $\nu = m/m(\{0\})$. This is a measurable normalization of a locally finite measure, with $\nu(\{0\}) = 1$, and its law remains dilation-invariant. For every compact interval K disjoint from zero,

$$(t \mapsto \lambda^n t)_* \nu(K) = \nu(\lambda^{-n} K) \longrightarrow 0.$$

Indeed, $\lambda^{-n} K$ lies in a shrinking punctured neighborhood of zero, whose mass tends to zero by local finiteness. Dilation invariance makes each quantity in this display have the same distribution as $\nu(K)$. Almost-sure convergence to zero therefore forces $\nu(K) = 0$ almost surely. A countable family of such intervals covers $\mathbb{R} \setminus \{0\}$, and hence $\nu = \delta_0$ almost surely. The atom event of probability zero is harmless. Combining this conclusion with the first part proves the lemma. $\square$

7 Activity along long diagonal segments

The root probabilities constructed above detect variation within a plaque at the intermediate resolutions. We now use the arithmetic tube estimate to show that this variation cannot disappear across a cut for a positive proportion of starting points over a long diagonal segment. A positive average under μ_i alone could coexist with zero activity on a set of starting points of positive measure. If $\mu_i(E_i) \geq p > 0$, the normalized restrictions $\xi_i = \mu_i|_{E_i}/\mu_i(E_i)$ satisfy $\xi_i \leq p^{-1}\mu_i$. Proving the estimate for every such dominated law excludes these positive-mass exceptional sets. In Section 8, this stronger conclusion will give almost-sure segment positivity for the limiting functions.

An *oriented cut* is a nonempty proper subset $J \subset \{1, \dots, 6\}$, with the roots directed from J to its complement:

$$\mathcal{E}_J = \{\beta_{jk} : j \in J, k \notin J\}.$$

The complementary subset gives the opposite orientation. Fix a small $\delta > 0$. For each such J , choose a regular $b_J \in L$ such that

$$\beta(b_J) \geq 1 \quad (\beta \in \mathcal{E}_J), \quad |(b_J)_k - (b_J)_j| \leq \delta \quad (j, k \text{ on the same side of } J). \quad (7.1)$$

These choices can be made in a fixed bounded subset of L : put the two sides at two distinct levels, perturb the entries within each side, and subtract their common average. There are only finitely many cuts. Write $\eta = 1/3$ for the exponential constant in Proposition 2.6; it is independent of δ . Write $g_{\beta,i} = g_{jk,i}$ when $\beta = \beta_{jk}$, and put

$$A_{J,i}(x) = \sum_{\beta \in \mathcal{E}_J} g_{\beta,i}(x).$$

Choose the common interior scale R_i as in (6.1), leaving separated interior depths both below and above it. In particular, $T_{-,i} \prec R_i \prec T_{+,i}$.

Proposition 7.1 (Activity on every long segment). *For δ sufficiently small, the preceding choices have the following property. For every oriented cut J , every $C < \infty$, every $\tau > 0$, and every sequence of probabilities ξ_i on X satisfying $\xi_i \leq C\mu_i$,*

$$\liminf_{i \rightarrow \infty} \frac{1}{R_i} \int_0^{\tau R_i} \int_X A_{J,i}(xa(tb_J)) d\xi_i(x) dt > 0. \quad (7.2)$$

The same assertion holds after passage to any subsequence.

The proof compares two bounds for the entropy of orbit names. The tube estimate gives a positive entropy cost per unit time. If the roots crossing the cut are inactive, the plaque probabilities vary only in directions whose expansion rates are at most δ . Their names then have a much smaller coding cost. This use of arithmetic separation to bound the entropy of diagonal orbit names belongs to the method of [8]; here the conditioning at intermediate resolutions and the arbitrary dominated starting probabilities require the additional details below. We use Shannon entropy with natural logarithms, including its chain rule and the elementary finite-alphabet error bound; see [2, Chapter 2].

7.1 Partitions and the cusp

For a countable partition $\mathcal{P}$ and a probability ν , write

$$H_\nu(\mathcal{P}) = - \sum_{P \in \mathcal{P}} \nu(P) \log \nu(P), \quad 0 \log 0 = 0.$$

Fix a regular $b \in L$ and put $T(x) = xa(b)$. The notation $\mathcal{P}_{[r,s]} = \bigvee_{t=r}^s T^{-t}\mathcal{P}$ records the symbols at the integer times from r through s . Let V be the pattern group of the b -positive roots.

Lemma 7.2 (A partition for consistent orbit names). *Given a sufficiently small identity neighborhood $O \subset G$, there is a countable Borel partition $\mathcal{P}$ of X with the following properties.*

1. *Every atom has μ -null boundary and lies in an inner part of a relatively compact VA plaque chart from the fixed stock. The chart extends beyond the atom with a margin.*
2. *If x, y have the same symbols at times $0, \dots, N$, there are relative lifts $h_t \in O$, $0 \leq t \leq N$, such that*

$$ya(tb) = xa(tb)h_t, \quad h_{t+1} = a(-b)h_t a(b).$$

3. *For every fixed C , the entropies $H_\nu(\mathcal{P})$ are bounded uniformly for all sufficiently large i and all probabilities $\nu \leq C\mu_i$. Moreover, there are finite coarsenings $\mathcal{P}^{(q)}$, obtained by keeping finitely many atoms and combining the others into one atom, for which*

$$\lim_{q \rightarrow \infty} \limsup_{i \rightarrow \infty} \sup_{\nu \leq C\mu_i} H_\nu(\mathcal{P} \mid \mathcal{P}^{(q)}) = 0. \quad (7.3)$$

Proof. Let $m(x)$ denote the shortest-vector length. Divide X into layers on which $m(x)$ is comparable to e^{-k} , for integers $k \geq 0$, using a first layer that also contains the thick part. Layer boundaries may be chosen μ -null. Successive minima and lattice basis reduction give a representative matrix g for each point of the k th layer such that

$$\|g\| + \|g^{-1}\| \leq e^{C_1(1+k)}.$$

Consequently the injectivity radius for relative group coordinates on that layer is at least $e^{-C_2(1+k)}$. Indeed, if a nonidentity integral matrix γ gave a smaller return, conjugating the return by g would give $\|\gamma - 1\| < 1$, which is impossible.

We impose a quantitative margin condition on these charts. Write d_G for the fixed left-invariant Riemannian distance on G . In a fixed identity neighborhood of G , adapted product coordinates give a smooth retraction π onto VA , with $\pi(h) = h$ for $h \in VA$ and

$$d_G(h, \pi(h)) \leq C_\pi \text{dist}_G(h, VA), \quad d_G(1, \pi(h)) \leq C_\pi d_G(1, h).$$

After shrinking the neighborhood, 1 and $\pi(h)$ are joined inside VA by a path of length at most $C_\pi d_G(1, h)$; one may use the local exponential coordinates of VA .

Choose outer VA product charts in the k th layer of radius comparable to $r_k = e^{-C_3(1+k)}$, with C_3 sufficiently large for injectivity, and inner boxes whose relative-coordinate distance from the outer boundary is at least a fixed multiple of r_k . Use translates of a fixed smooth group-coordinate chart, so that its local product and metric comparison constants are uniform. Cover the layer by still smaller symbol boxes inside these inner boxes. Require their relative-group diameter to be at most $c_* r_k$, where $c_* > 0$ is small enough compared with both the inner-to-outer margin and the radius of the domain of π . In particular, whenever $x, y = xh$ are in one symbol box, the whole path from x to $x\pi(h)$ just described stays inside its outer chart. It follows that $x\pi(h)$ is on the same chart plaque as x , rather than on another local sheet of that orbit.

The number of symbol boxes needed is at most $e^{C_4(1+k)}$. Indeed, subdivide the bounded-matrix representatives at an exponentially small mesh, and refine by the fixed factor c_* when necessary. These operations preserve an exponential bound in k . Passing from the cover to a partition by successive differences does not increase the number of pieces or their diameters. The charts and

their cuts can be chosen from the stock with μ -null boundaries: perturb centers within a small fixed fraction of the covering radius and choose each coordinate width in a fixed comparable interval, avoiding the at most countably many widths that give a boundary of positive measure. These choices preserve the exponential bounds on sizes and on the number of boxes, as well as the inner-chart margins. Thus each partition atom remains inside a chart with a margin, although it need not itself be a coordinate rectangle.

The shortest-vector length changes by at most a fixed multiplicative factor under T . We can therefore choose the boxes so small that both a relative lift in one symbol and its conjugate by $a(b)$ lie in an injective neighborhood at the next symbol. Uniqueness in that neighborhood identifies the latter lift with the lift chosen there. Induction gives the consistency in assertion 2. Decreasing all box sizes by a common fixed factor ensures that every lift lies in the prescribed O .

By Proposition 2.3, a probability $\nu \leq C\mu_i$ assigns the k th layer mass at most C_5e^{-ck} , uniformly for sufficiently large i . If that mass is p_k and there are at most M_k symbols in the layer, their entropy is bounded by

$$p_k \log M_k - p_k \log p_k.$$

Since $\log M_k = O(1 + k)$ and $p_k \leq C_5e^{-ck}$, the sum of these bounds converges uniformly, and its tail tends to zero uniformly. Keeping all symbols from the first finitely many layers proves both entropy assertions. $\square$

7.2 What an inactive cut implies inside a cell

Fix J , put $b = b_J$, and let V_s be the pattern of positive roots whose endpoints lie on the same side of J . Set $H_s = V_s A$. The cross-root group, denoted Z_J , is normal in VA , and each element of VA has a unique factorization

$$h = zd, \quad z \in Z_J, \quad d \in H_s.$$

In a product chart $x = \psi(q)zd$, the H_s plaques are precisely the sets with (q, z) fixed. Lemma 6.2 says that at a limiting state whose crossing-root indicators all vanish, the VA probability is supported on the H_s plaque through the sampled point.

We need this support assertion for conditional probabilities of a dominated measure, rather than only for μ_i . The following consequence records the required change of measure.

Lemma 7.3. *Fix finitely many atoms S of $\mathcal{P}$, with their VA charts, and an interior transverse depth $l_i \prec R_i$. Suppose that probabilities $\nu_i \leq C\mu_i$ satisfy $\int A_{J,i} d\nu_i \rightarrow 0$. Condition on one of these symbols S and on its transverse q cell. Draw two points independently from the resulting conditional probability of ν_i . Then their z coordinates differ by a quantity tending to zero in probability, averaged with the probabilities of the conditioning events.*

The conclusion also holds on average over any finite list of such probabilities, with arbitrary probability weights, if their weighted mean of $\int A_{J,i} d\nu_i$ tends to zero.

Proof. First sample x under ν_i , and draw the second plaque coordinate from the μ_i probability conditioned only on its transverse cell in the whole chart. The depth-comparison rules allow this conditional probability to be replaced, in the limit, by the chart probability in the state data. Any joint limit of the state with x sampled under ν_i is dominated by C times the usual state law. Its crossing-root indicators vanish almost surely, since their nonnegative sum has expectation tending to zero. Lemma 6.2 therefore puts the second point on the same H_s plaque as the sample. Its z coordinate equals the sampled z . The transverse mesh tends to zero; continuous coordinate tests on the fixed compact chart boxes now show that the prelimit z difference tends to zero in probability.

This argument uses the probability attached to one chart and its statewise support. It does not condition the sampled point on the other plaque probabilities in the state.

To change the second draw to the stated ν_i conditional law, fix $\varepsilon > 0$ and discard events $E = S \cap \{q \text{ in a given cell}\}$ such that

$$\nu_i(E) < \varepsilon \mu_i(\{q \text{ in that cell}\} \cap \text{chart}).$$

Their total ν_i mass is at most ε times the number of symbols under consideration. On every remaining event the conditional ν_i law has density at most C/ε with respect to the μ_i chart-cell law. The preceding convergence thus remains valid for the second draw. Let $i \rightarrow \infty$ and then $\varepsilon \rightarrow 0$.

For the averaged statement, first observe a uniform sequential consequence of the first assertion: no sequence of dominated probabilities can have indicator expectation tending to zero while the asserted pair error stays bounded away from zero. Let $a_i \rightarrow 0$ be the weighted mean of the indicator expectations in a list. Discard members whose expectation is greater than $\sqrt{a_i} + i^{-1}$. Their total weight is at most $a_i/\sqrt{a_i} + i^{-1} \rightarrow 0$ by Markov's inequality. On the remaining members, the pair errors tend uniformly to zero by the sequential consequence: otherwise a sequence of members violating it could be selected. This proves the averaged assertion. $\square$

7.3 The lower and upper entropy bounds

Proof of Proposition 7.1. Suppose, along a subsequence, that the expression in (7.2) tends to zero. Choose even integers $N_i \sim \tau R_i$ with $N_i + 2 < \tau R_i$. Averaging a starting shift $s \in [0, 1]$ shows that we may replace ξ_i by its translate through some $s_i \in [0, 1]$ so that

$$\frac{1}{N_i + 1} \sum_{t=0}^{N_i} \int A_{J,i}(T^t x) d\xi_i(x) \rightarrow 0. \quad (7.4)$$

All these translated probabilities still satisfy $\xi_i \leq C\mu_i$. More generally, every time- t distribution $\nu_{i,t} = (T^t)_* \xi_i$ has the same domination bound.

Choose a compact set K with $\nu_{i,N_i/2}(K) \geq 1 - \varepsilon_0$ for all sufficiently large i , where $\varepsilon_0 > 0$ will be small. This follows from the uniform cusp estimate and domination. Take the neighborhood O in Lemma 7.2 small enough to use Proposition 2.6 uniformly on K . If a name atom $B \in \mathcal{P}_{[0,N_i]}$ meets $T^{-N_i/2}K$, choose a reference point in that intersection. Consistency of the relative lifts puts every midpoint of B in the tube about its reference midpoint with time $N_i/2$. Since $T_{-,i} \prec R_i \prec T_{+,i}$, this time is in the arithmetic range for all sufficiently large i . Hence

$$\xi_i(B) \leq C e^{-\eta N_i/2}.$$

Such atoms have total ξ_i mass at least $1 - \varepsilon_0$. Integrating the negative logarithm of the atom mass gives, after decreasing ε_0 and taking i large,

$$\frac{1}{N_i} H_{\xi_i}(\mathcal{P}_{[0,N_i]}) \geq \frac{\eta}{3}. \quad (7.5)$$

We next derive the incompatible upper bound. Fix a small target error $\varepsilon > 0$. First choose a finite coarsening $\mathcal{P}^0$ so that the conditional entropy in (7.3) is as small as required below. Let q_0 be its alphabet size. Choose also a finite collection $\mathcal{S}$ of original symbols whose total complement has arbitrarily small probability under every $\nu \leq C\mu_i$. The symbols in $\mathcal{S}$ need not be the symbols retained in $\mathcal{P}^0$.

The boundary of $\mathcal{P}^0$ is a finite union of compact μ -null sets. We can choose a fixed distance accuracy $\rho > 0$ so that its 2ρ neighborhood has arbitrarily small probability under every $\nu \leq C\mu_i$

for all sufficiently large i . To justify this uniformly, take decreasing closed neighborhoods of these boundaries; their μ masses tend to zero, and the weak convergence upper bound applies to each fixed neighborhood. All subsequent orbit predictions will be within distance ρ of the actual point. Incorrect symbols can then occur only near these boundaries, or on the exceptions specified below.

Choose transverse grids in the charts for $\mathcal{S}$ at an interior depth $l_i \prec R_i$. Let M be a large integer, fixed while $i \rightarrow \infty$. Its size will be chosen after the charts and ρ ; the concentration tolerances used below may then depend on M . Split the name into blocks of M future symbols given the past through the block's initial time. We may omit an initial $t_i = o(N_i)$ symbols and a final remainder of length less than M : their entropy cost divided by N_i tends to zero by the uniform single-symbol entropy bound.

Here is why the extra transverse-cell label costs only a bounded amount at every remaining block. In a past-name atom at time t , the negative root entries of its relative lift are $O(e^{-c_b t})$, where $c_b = \min_{\beta(b) > 0} \beta(b) > 0$. Triangular coordinates in a small group neighborhood therefore put the lift at distance $O(e^{-c_b t})$ from VA . Let $x, y = xh$ be two current points in one past-name atom, and use the retraction π from the partition construction. The quantitative diameter condition ensures that $x\pi(h)$ lies on the same plaque of the current outer chart as x . Thus $q(x\pi(h)) = q(x)$. Lipschitz continuity of the transverse coordinate on that chart gives

$$\begin{aligned} |q(y) - q(x)| &= |q(xh) - q(x\pi(h))| \\ &\leq C d_G(h, \pi(h)) \leq C' \text{dist}_G(h, VA) \leq C'' e^{-c_b t}. \end{aligned}$$

The constants are uniform over the fixed finite collection of current charts. Choose t_i to be a sufficiently large fixed multiple of l_i ; then $e^{-c_b t} < 2^{-l_i}$ for $t \geq t_i$, and $t_i = o(N_i)$. Consequently the possible transverse coordinates meet only a bounded number B_0 of neighboring cells. Given the past, their label has entropy at most $\log B_0$. For current symbols outside $\mathcal{S}$ use a single dummy label.

Let $\mathcal{Q}_i$ record this cell label in the current symbol, with the dummy label outside $\mathcal{S}$, and write $\mathcal{F}_t = \mathcal{P}_{[0,t]}$. After telling the cell label, we may discard the rest of the past when upper-bounding the entropy of the next coarsened block. Precisely, the chain rule and monotonicity under conditioning give

$$\begin{aligned} H_{\xi_i}((\mathcal{P}^0)_{[t+1, t+M]} \mid \mathcal{F}_t) \\ &\leq H_{\xi_i}(T^{-t} \mathcal{Q}_i \mid \mathcal{F}_t) + H_{\nu_{i,t}}((\mathcal{P}^0)_{[1,M]} \mid \mathcal{P} \vee \mathcal{Q}_i) \\ &\leq \log B_0 + H_{\nu_{i,t}}((\mathcal{P}^0)_{[1,M]} \mid \mathcal{P} \vee \mathcal{Q}_i). \end{aligned} \tag{7.6}$$

The conditioning in the last entropy is just the current symbol and its cell. By (7.4), the mean crossing-root activity at the block starts tends to zero. Indeed their nonnegative sum is bounded by the full sum in that equation, and normalizing by the number of blocks loses at most the fixed factor M . The averaged form of Lemma 7.3 applies to their distributions. For each symbol-cell event we may therefore choose a z center so that, apart from a probability tending to zero on average, z lies within any prescribed fixed tolerance of that center. Such a center exists by averaging the two-draw error and choosing one of the points as center.

The d coordinates lie in a fixed compact part of H_s for each of the finitely many charts. Every root occurring in H_s has expansion rate at most δ , and A commutes with the motion. A mesh in these coordinates, of accuracy a fixed multiple of $\rho e^{-\delta M}$, therefore predicts the trajectory through the next M steps to the required distance accuracy, provided the z and transverse coordinates are sufficiently close to their centers. The number of mesh points is at most

$$C_0 \exp(C_6 \delta M). \tag{7.7}$$

Here C_6 depends only on the dimension, while C_0 depends on the fixed charts and ρ , not on M . This follows by using finitely many smooth coordinate meshes on the compact d ranges: conjugation expands every entry of a group-relative error by at most $e^{\delta M}$. Left invariance of the metric transfers the resulting relative-group error to an error on X .

The cross-root center tolerance may need to be exponentially small in M , because cross-root directions expand faster. This causes no additional mesh labels: M is fixed before applying the concentration lemma, so any such positive tolerance is allowed. The transverse width also tends to zero for each fixed M . Thus C_0 in (7.7) remains independent of M .

Encode the next M symbols of $\mathcal{P}^0$ by a mesh label, with a single additional label for an omitted chart or an outlier, and then by corrections to the predicted symbols. Let $e_{i,t}$ be the average probability of an incorrect prediction over these M positions, with the starting point sampled under $\nu_{i,t}$. Put $h_2(e) = -e \log e - (1 - e) \log(1 - e)$, with the usual zero conventions. The mesh and correction costs give

$$\begin{aligned} H_{\nu_{i,t}}((\mathcal{P}^0)_{[1,M]} \mid \mathcal{P} \vee \mathcal{Q}_i) \\ \leq \log(C_0 e^{C_6 \delta M} + 1) + M(h_2(e_{i,t}) + e_{i,t} \log q_0). \end{aligned} \quad (7.8)$$

For completeness, reveal the error indicator at each position and, on an error, reveal the actual symbol; subadditivity and concavity of h_2 give this bound also when the error probabilities differ across positions and blocks. The chart omissions and boundary neighborhoods were chosen with arbitrarily small probabilities. The additional concentration errors tend to zero on average. Consequently the mean of $e_{i,t}$ over the block starts can be made as small as required; concavity of h_2 gives the same small correction bound after averaging blocks.

Finally, the original block is recovered from its coarsened block at the following cost:

$$\begin{aligned} H_{\xi_i}(\mathcal{P}_{[t+1,t+M]} \mid \mathcal{F}_t) \\ \leq H_{\xi_i}((\mathcal{P}^0)_{[t+1,t+M]} \mid \mathcal{F}_t) + \sum_{r=1}^M H_{\nu_{i,t+r}}(\mathcal{P} \mid \mathcal{P}^0). \end{aligned} \quad (7.9)$$

Indeed, conditional on the entire coarsened block, revealing each original symbol costs at most its one-step conditional entropy given its own coarsened symbol. Every summand on the right is controlled by (7.3), because $\nu_{i,t+r} \leq C\mu_i$.

Sum (7.9) over the blocks and use (7.6) and (7.8). The chain rule, together with the negligible initial and final pieces, now gives

$$\limsup_{i \rightarrow \infty} \frac{1}{N_i} H_{\xi_i}(\mathcal{P}_{[0,N_i]}) \leq C_6 \delta + \varepsilon + \frac{\log B_0 + \log(C_0 + 1)}{M}. \quad (7.10)$$

All constants in the last numerator were fixed before M was chosen. Choose δ first so that $C_6 \delta < \eta/12$, then choose the truncation, charts and distance accuracy to make $\varepsilon < \eta/12$, and finally choose M so large that the last term is less than $\eta/12$. Letting $i \rightarrow \infty$ contradicts (7.5). This proves the proposition, including its subsequence formulation. $\square$

8 Combining the root orientations

Proposition 7.1 guarantees activity somewhere on every long segment across each oriented cut. It does not yet assert that the roots are active at the same point. The wall invariance of the root indicators supplies the missing relation: on the scale R_i , an indicator can be read as a function of

the value of its root alone. Transitivity then becomes an approximate addition rule for functions on the real line. We first obtain activity in every real interval and then use two-edge cycles to show that the inactive sets have measure zero.

Theorem 8.1 (Full activity). *For every ordered pair $j \neq k$ in $\{1, \dots, 6\}$,*

$$\int_X g_{jk,i} d\mu_i \longrightarrow 1. \quad (8.1)$$

Consequently every root is active almost surely in the limiting state law.

8.1 Functions of a root coordinate

For each root β , fix $b_\beta \in L$ such that $\beta(b_\beta) = 1$. If x is sampled with law μ_i , define the random measurable functions

$$G_{\beta,i}^x(s) = g_{\beta,i}(xa(R_i s b_\beta)), \quad s \in \mathbb{R}. \quad (8.2)$$

They take values in $\{0, 1\}$. Regard them as elements of the compact metrizable space

$$\mathcal{B} = \{f \in L_{\text{loc}}^\infty(\mathbb{R}) : 0 \leq f \leq 1\},$$

with weak-* convergence on every bounded interval. More explicitly, one may use a countable dense set of L^1 test functions on each $[-q, q]$, $q \in \mathbb{N}$, to metrize this topology. The finite product $\mathcal{B}^{30}$ is also compact and metrizable. After passing to a subsequence, the laws of the arrays in (8.2) converge to a law of functions $(F_\beta)_\beta$ in this space. We also write $G_{jk,i}^x = G_{\beta_{jk},i}^x$ and $F_{jk} = F_{\beta_{jk}}$.

Lemma 8.2. *For any such convergent subsequence, the limiting law and its prelimit arrays have the following properties.*

1. *Almost surely, for every $v_0 \in L$ with rational coordinates, every oriented cut J , and every positive rational τ ,*

$$\sum_{\beta \in \mathcal{E}_J} \int_0^\tau F_\beta(\beta(v_0 + tb_J)) dt > 0. \quad (8.3)$$

2. *For distinct j, k, ℓ , and every bounded rectangle $B \subset \mathbb{R}^2$, the nonnegative defects satisfy*

$$\mathbb{E}_{x \sim \mu_i} \int_B G_{jk,i}^x(s) G_{k\ell,i}^x(t) (1 - G_{j\ell,i}^x(s+t)) ds dt \longrightarrow 0. \quad (8.4)$$

Proof. For v in a fixed bounded subset of L , the vector $v - \beta(v)b_\beta$ belongs to $\ker \beta$ and remains bounded. Wall invariance (6.1), followed by invariance of μ_i under the common translation, gives

$$\sup_{v \text{ in a fixed bounded set}} \int_X \left| G_{\beta,i}^x(\beta(v)) - g_{\beta,i}(xa(R_i v)) \right| d\mu_i(x) \longrightarrow 0. \quad (8.5)$$

Fix the parameters in assertion 1. The left side of (8.3) is a continuous function of the weak-* array, because $\beta(b_J) \neq 0$ for every crossing root. By (8.5), its prelimit differs in $L^1(\mu_i)$ by $o(1)$ from

$$Y_i(x) = \sum_{\beta \in \mathcal{E}_J} \int_0^\tau g_{\beta,i}(xa(R_i(v_0 + tb_J))) dt.$$

Suppose that the limiting observable were zero on a set of probability $p > 0$. Convergence in law and nonnegativity allow us to choose thresholds $\varepsilon_i \downarrow 0$ and events E_i with $\mu_i(E_i) \geq p/2$ such that the

original prelimit observable is at most ε_i on E_i . To see this directly, first choose a fixed sufficiently small continuity threshold, use convergence in law, and then decrease the threshold along a diagonal subsequence. The normalized restrictions $\mu_i|_{E_i}/\mu_i(E_i)$ consequently have Y_i expectation tending to zero. Translate these probabilities by $a(R_i v_0)$ and call the results ξ_i . They satisfy $\xi_i \leq (2/p)\mu_i$, whereas the change of variable $u = R_i t$ gives

$$\frac{1}{R_i} \int_0^{\tau R_i} \int_X A_{J,i}(xa(ub_J)) d\xi_i(x) du \longrightarrow 0.$$

This contradicts Proposition 7.1. The parameters in assertion 1 range over a countable set, so the assertion holds simultaneously almost surely.

For assertion 2, the roots β_{jk} and $\beta_{k\ell}$ are linearly independent. Choose $v(s, t)$ linearly in (s, t) with

$$\beta_{jk}(v(s, t)) = s, \quad \beta_{k\ell}(v(s, t)) = t.$$

It follows that $\beta_{j\ell}(v(s, t)) = s + t$. On a bounded rectangle, (8.5) permits replacing all three functions in the defect by their indicators at the common point $xa(R_i v(s, t))$, with total expected error $o(1)$. The expected defect there is $o(1)$ by the approximate transitivity in Lemma 6.3; its distribution is unchanged by the translation. Integrating over the rectangle proves the claim. $\square$

The transitivity defect is nonlinear in the functions, so we do not pass it directly through weak-* convergence. Instead we select deterministic arrays for which the defects themselves tend to zero.

Lemma 8.3. *If (8.1) fails for some root, there is a subsequence and a deterministic choice of arrays $(G_{\beta,i})_\beta$, each taken from (8.2), with the following properties:*

1. $G_{\beta,i} \rightarrow F_\beta$ weak-* on bounded intervals, and the deterministic limiting functions satisfy (8.3) for all its stated parameters;
2. every defect in (8.4), without its expectation, tends to zero on every bounded rectangle;
3. for some root β_0 and some bounded interval I of positive length,

$$\int_I (1 - F_{\beta_0}(s)) ds > 0. \tag{8.6}$$

Proof. Pass to a subsequence on which $\int_X (1 - g_{\beta_0,i}) d\mu_i \geq \varepsilon$ for a fixed $\varepsilon > 0$. Invariance of μ_i and Fubini's theorem imply, for any fixed interval I ,

$$\mathbb{E} \int_I (1 - G_{\beta_0,i}^x(s)) ds = |I| \int_X (1 - g_{\beta_0,i}) d\mu_i \geq \varepsilon |I|.$$

The integral is a bounded weak-* continuous observable, so the same inequality holds for the limiting law. This law therefore gives positive probability to a positive deficiency. Choose a point F in its support that has a positive deficiency and satisfies assertion 1 of Lemma 8.2.

For each integer q , choose a neighborhood U_q of F of diameter less than $1/q$. It has positive limiting probability, and thus the prelimit probabilities of U_q are bounded below by a positive number for all sufficiently large indices. There are finitely many root compositions. By (8.4), the sum of their defects on $[-q, q]^2$ has expectation tending to zero. Choose an index so large that, by Markov's inequality, the probability that this sum exceeds $1/q$ is less than half the probability of U_q . There is therefore a sampled array in U_q with total defect at most $1/q$. Make these choices with increasing indices. They converge to F and have vanishing defects on every bounded rectangle, by nonnegativity and containment in some $[-q, q]^2$. $\square$

8.2 From cut activity to every interval

For the rest of the argument, fix deterministic arrays as in Lemma 8.3. The labels are still $\{0, 1\}$ -valued; their weak-* limits may initially take any value in $[0, 1]$.

Lemma 8.4. *For every root β and every nonempty open interval $I \subset \mathbb{R}$,*

$$\int_I F_\beta(s) ds > 0.$$

Proof. Fix a root β_{jk} and a desired interval I ; replacing it by a bounded nonempty open subinterval only strengthens what we must prove. Choose a rational $v_0 \in I$ so that $\beta_{jk}(v_0)$ lies in I . Choose $r > 0$ small enough that

$$[\beta_{jk}(v_0) - 6r, \beta_{jk}(v_0) + 6r] \subset I.$$

For every root α , put $I_\alpha = (\alpha(v_0) - r, \alpha(v_0) + r)$. Draw the directed edge $p \rightarrow q$ if $\int_{I_{\beta_{pq}}} F_{pq} > 0$.

This directed graph is strongly connected. Indeed, for any nonempty proper subset J , choose a positive rational τ sufficiently small that $\alpha(v_0 + tb_J) \in I_\alpha$ for every $\alpha \in \mathcal{E}_J$ and $0 \leq t \leq \tau$. Equation (8.3) then supplies an edge from J to its complement. A finite directed graph with an outgoing edge across every nontrivial cut is strongly connected: otherwise the set of vertices reachable from a suitable vertex is a nonempty proper set with no outgoing edge.

Take a simple directed path $j = p_0, p_1, \dots, p_m = k$, where $m \leq 5$. Write $\alpha_h = \beta_{p_{h-1}p_h}$ and $I_h = I_{\alpha_h}$. Weak-* convergence gives positive lower bounds, for sufficiently large i , on

$$\int_{I_h} G_{\alpha_h, i}(s) ds, \quad 1 \leq h \leq m.$$

The product of these integrals is consequently bounded below by a positive constant. Interpret it as the measure of tuples $(s_1, \dots, s_m) \in \prod_h I_h$ on which all the path labels equal one.

Except for a set of tuple measure tending to zero, successive composition of these labels gives

$$G_{p_0 p_h, i}(s_1 + \dots + s_h) = 1, \quad 1 \leq h \leq m.$$

Here is the integration detail. At step h , the failure of this implication, if all earlier implications held, is bounded by the transitivity defect for p_0, p_{h-1}, p_h . The path is simple, so these are distinct vertices. Integration in $s_1, \dots, s_{h-1}$ pushes Lebesgue measure on bounded intervals to a measure of bounded density in their sum. Multiplying by previous indicator restrictions can only decrease this measure. The defect integral on a fixed bounded rectangle therefore bounds the failure measure by a fixed multiple of an $o(1)$ quantity. Summing over the finitely many steps proves the assertion. If $m = 1$, no composition is necessary.

The final sum $s_1 + \dots + s_m$ lies within mr of $\alpha_1(v_0) + \dots + \alpha_m(v_0) = \beta_{jk}(v_0)$, hence in I . Its pushforward density is bounded above by a fixed constant. Since a positive measure of tuples has final label one, we obtain $\liminf_i \int_I G_{jk, i} > 0$. Weak-* convergence proves the lemma. $\square$

We have now located positive activity in every interval for every root. To eliminate a positive inactive set, we will translate an active set by sums of parameters from two opposite auxiliary roots. Composing successively with those two roots returns to the original root while adding their two parameters. Interval positivity supplies parameter sets of positive measure on both sides of zero. By mixing normalized Lebesgue measure on these sets, we will obtain bounded densities of mean zero, bounded support, and variance bounded away from zero. The sum of two independent parameters with these densities describes one cycle increment. For a fixed bounded interval I , the argument needs a fixed number of independent cycle increments to have density bounded below on $I - I$: this makes every part of the active set interact with every part of the inactive set inside I . The next lemma gives that lower bound uniformly over the densities that arise.

8.3 A uniform convolution estimate

Lemma 8.5. Fix $B, M, v, L > 0$. Consider the probability densities p on $\mathbb{R}$ such that

$$\text{supp } p \subset [-B, B], \quad 0 \leq p \leq M, \quad \int xp(x) dx = 0, \quad \int x^2 p(x) dx \geq v.$$

There are an integer $m \geq 2$ and $c > 0$, depending only on B, M, v, L , such that

$$p^{*m}(x) \geq c \quad (|x| \leq L)$$

for every such density. The inequality uses the continuous representative of the convolution density.

Proof. If the family is empty there is nothing to prove. Otherwise write $\sigma_p^2 = \int x^2 p(x) dx$, so $v \leq \sigma_p^2 \leq B^2$, and let

$$\phi_p(t) = \int e^{itx} p(x) dx.$$

Bounded support and the zero mean give the uniform expansion

$$\phi_p(t) = 1 - \frac{1}{2} \sigma_p^2 t^2 + O(B^3 |t|^3). \quad (8.7)$$

Applying the same expansion to $|\phi_p(t)|^2$, the characteristic function of the difference of two independent variables with density p , shows that for a sufficiently small $\delta > 0$, depending only on B, v ,

$$|\phi_p(t)| \leq e^{-vt^2/4} \quad (|t| \leq \delta). \quad (8.8)$$

There is also a uniform gap below one outside this neighborhood. For $|t| \geq \delta$, an angle θ , and $0 < a < \pi$, the set

$$\{x \in [-B, B] : \text{dist}(tx - \theta, 2\pi\mathbb{Z}) < a\}$$

has Lebesgue measure at most $a(2B/\pi + 4/\delta)$. Indeed, each complete period contributes the fraction a/π of its length, and at most two incomplete periods contribute at the ends. Choose a so small that M times this bound is at most $1/2$. With $\theta = \arg \phi_p(t)$, at least half of the probability lies outside this arc. Taking the real part after rotation by $e^{-i\theta}$ gives

$$\sup_p \sup_{|t| \geq \delta} |\phi_p(t)| \leq \frac{1 + \cos a}{2} =: \rho < 1. \quad (8.9)$$

If $\phi_p(t) = 0$, the inequality is immediate and the angle is irrelevant. Notice that the estimate holds at arbitrarily large frequencies; no uniform version of the Riemann–Lebesgue lemma is needed.

Plancherel's identity yields

$$\int_{\mathbb{R}} |\phi_p(t)|^2 dt = 2\pi \int_{\mathbb{R}} p(x)^2 dx \leq 2\pi M.$$

For $m \geq 2$, ϕ_p^m is integrable. Fourier inversion therefore defines a continuous representative of p^{*m} . We claim the following uniform local central limit estimate:

$$\sup_p \sup_{x \in \mathbb{R}} \left| \sqrt{m} p^{*m}(x) - \frac{1}{\sqrt{2\pi} \sigma_p} \exp\left(-\frac{x^2}{2m\sigma_p^2}\right) \right| \rightarrow 0. \quad (8.10)$$

To verify it, change variables $u = t\sqrt{m}$ in the inversion formula. The difference in (8.10) is at most $1/(2\pi)$ times

$$\int_{\mathbb{R}} \left| \phi_p(u/\sqrt{m})^m - e^{-\sigma_p^2 u^2/2} \right| du.$$

On each fixed bounded u interval, the expansion (8.7) proves uniform convergence of the integrand to zero; the logarithm may be taken near one, and its error is $O(B^3|u|^3/\sqrt{m})$. On $|u| \leq \delta\sqrt{m}$, (8.8) bounds the first term by $e^{-vu^2/4}$, while the Gaussian term is bounded by $e^{-vu^2/2}$. Thus the contribution outside a large fixed bounded interval but inside this region is uniformly small. On $|u| > \delta\sqrt{m}$, (8.9) and Plancherel give

$$\int_{|u|>\delta\sqrt{m}} |\phi_p(u/\sqrt{m})|^m du \leq 2\pi M\sqrt{m} \rho^{m-2} \rightarrow 0.$$

The Gaussian tail there also tends to zero uniformly. These three estimates prove (8.10).

For $|x| \leq L$ and all sufficiently large m , the Gaussian term in (8.10) is at least $1/(2\sqrt{2\pi}B)$, uniformly in p . Increase m so that the error is at most $1/(4\sqrt{2\pi}B)$. Then

$$p^{*m}(x) \geq \frac{1}{4\sqrt{2\pi}B\sqrt{m}} \quad (|x| \leq L),$$

which is the required conclusion. $\square$

8.4 Cycles remove the inactive sets

Proof of Theorem 8.1. Suppose the theorem fails, and take the deterministic arrays and limiting functions from Lemma 8.3. Fix an edge $j \rightarrow k$, and choose ℓ distinct from j, k . Lemma 8.4 supplies, for each of the roots ℓj and $j\ell$, positive limiting integral on each of the intervals $(-2, -1)$ and $(1, 2)$. Accordingly, for all sufficiently large i , each of the four sets where the relevant G equals one has length bounded below by a positive constant.

For each auxiliary root, normalize Lebesgue measure on its two sets to probability densities p_i^- and p_i^+ . Their means belong to $[-2, -1]$ and $[1, 2]$, respectively. There is a convex combination

$$p_i = \lambda_i p_i^- + (1 - \lambda_i) p_i^+$$

with mean zero. The coefficients can be chosen in $[1/3, 2/3]$ by solving the mean-zero equation. Thus these densities have support in $[-2, 2]$, a common finite upper bound, and variance at least one. They are supported entirely where the corresponding auxiliary indicator is one. Denote the density for ℓj by p_i and that for $j\ell$ by q_i .

Starting from $G_{jk,i}(t) = 1$, composition with a parameter s in the support of p_i gives the edge ℓk , and composition with a parameter s' in the support of q_i returns to jk :

$$\ell j + jk = \ell k, \quad j\ell + \ell k = jk.$$

Both are root compositions with three distinct vertices. Their defect estimates imply, for every fixed bounded interval I ,

$$\int_I \int_{\mathbb{R}} \int_{\mathbb{R}} G_{jk,i}(t) (1 - G_{jk,i}(t + s + s')) p_i(s) q_i(s') ds ds' dt \rightarrow 0. \quad (8.11)$$

Indeed, except on null sets of parameters, both auxiliary labels are one. The indicator in (8.11) is then bounded by the sum of the two composition defects. For the first defect, integrate out s' and use the uniform density bound on p_i . For the second, the variable $t + s$ has a bounded density on a fixed bounded interval when $t \in I$ is integrated with Lebesgue measure and s with p_i . The bound on q_i and the defect estimate for $(j\ell, \ell k)$ finish the verification.

Put $r_i = p_i * q_i$. These cycle-increment densities have support in $[-4, 4]$, a common density bound, mean zero, and variance at least two. We can iterate (8.11) any fixed number m of times to get

$$\int_I \int_{\mathbb{R}} G_{jk,i}(t)(1 - G_{jk,i}(t+u))r_i^{*m}(u) du dt \longrightarrow 0. \quad (8.12)$$

For clarity, along a sequence of m increments, a path starting with label one and ending with label zero has a first transition from one to zero. Bound its indicator by the sum of the indicators of these transitions. The intermediate positions lie in $I + [-4m, 4m]$. After integrating all preceding increments, their distributions have densities bounded by one relative to Lebesgue measure on that larger interval, because the initial t is integrated over I and all increments have probability laws. Each term is therefore bounded by the one-cycle error on the larger interval. There are only m terms, proving (8.12).

Fix a bounded interval I of positive length and apply Lemma 8.5 to the r_i , with L large enough that $I - I \subset [-L, L]$. Choose its fixed power m and lower bound $c > 0$. In (8.12), retain only the pairs $t, t+u \in I$. We obtain

$$c \left(\int_I G_{jk,i}(t) dt \right) \left(\int_I (1 - G_{jk,i}(t)) dt \right) \longrightarrow 0.$$

The first factor has positive lower limit by Lemma 8.4. Hence $\int_I (1 - G_{jk,i}) \rightarrow 0$ for every such interval. Weak-* convergence shows that $F_{jk} = 1$ almost everywhere. The argument applies to every root, contradicting the deficiency (8.6). Thus (8.1) holds.

By Lemma 6.3, the sequence indicators converge jointly in law to the activity indicators of the limiting state. Since each has expectation tending to one, every limiting root indicator equals one almost surely. There are only thirty roots, so this holds for all of them simultaneously. $\square$

9 Proof of the equidistribution theorem

We now return to the packets of Theorem 1.1. Proposition 2.3 and Mahler's compactness criterion give tightness. Take an arbitrary subsequence with a weak probability limit μ . It is A -invariant, and Proposition 2.4 gives the ordinary-ball estimate required by Proposition 3.1. Theorem 3.2 therefore makes almost every A -ergodic component of μ homogeneous.

Apply the intermediate-conditioning construction to this subsequence, passing further to a subsequence wherever necessary. The augmented limiting state still projects to μ on X . Proposition 2.6 supplies the tube estimate throughout the required time interval. Hence Theorem 8.1 makes every root active almost surely in this state.

If proper homogeneous components had positive total weight, Lemma 3.4 would place positive μ -mass on the countable union of their fixed-point pieces. At least one piece would have positive mass. Its fixing diagonal is nonidentity, so it has two unequal diagonal entries. The corresponding root plaques meet that piece in at most countably many points. By Lemma 6.4, that root is inactive almost surely on states projecting to the piece, a contradiction. Thus almost every component is the G -invariant probability, and $\mu = m_6$.

Every weakly convergent subsequence supplied by tightness has the same limit. It follows that the original sequence converges to m_6 on $C_c(X_6)$. Proposition 2.3 already gives the compact sets required for the nonescape assertion. The estimates are uniform in the embedding ordering, so the conclusion holds for the arbitrary choices in Theorem 1.1.

References

- [1] J. W. S. Cassels. *An Introduction to the Geometry of Numbers*, volume 99 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin–Göttingen–Heidelberg, 1959.
- [2] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, NJ, second edition, 2006. doi: 10.1002/047174882X.
- [3] William Duke. Hyperbolic distribution problems and half-integral weight Maass forms. *Invent. Math.*, 92(1):73–90, 1988. doi: 10.1007/BF01393993.
- [4] Manfred Einsiedler and Anatole Katok. Invariant measures on G/Γ for split simple Lie groups G . *Comm. Pure Appl. Math.*, 56(8):1184–1221, 2003. doi: 10.1002/cpa.10092.
- [5] Manfred Einsiedler and Anatole Katok. Rigidity of measures—the high entropy case and non-commuting foliations. *Israel J. Math.*, 148:169–238, 2005. doi: 10.1007/BF02775436.
- [6] Manfred Einsiedler and Elon Lindenstrauss. Diagonal actions on locally homogeneous spaces. In *Homogeneous Flows, Moduli Spaces and Arithmetic*, volume 10 of *Clay Mathematics Proceedings*, pages 155–241. American Mathematical Society, Providence, RI, 2010.
- [7] Manfred Einsiedler, Anatole Katok, and Elon Lindenstrauss. Invariant measures and the set of exceptions to Littlewood’s conjecture. *Ann. of Math. (2)*, 164(2):513–560, 2006. doi: 10.4007/annals.2006.164.513.
- [8] Manfred Einsiedler, Elon Lindenstrauss, Philippe Michel, and Akshay Venkatesh. Distribution of periodic torus orbits on homogeneous spaces. *Duke Math. J.*, 148(1):119–174, 2009. doi: 10.1215/00127094-2009-023.
- [9] Manfred Einsiedler, Elon Lindenstrauss, Philippe Michel, and Akshay Venkatesh. Distribution of periodic torus orbits and Duke’s theorem for cubic fields. *Ann. of Math. (2)*, 173(2):815–885, 2011. doi: 10.4007/annals.2011.173.2.5.
- [10] Manfred Einsiedler, Elon Lindenstrauss, Philippe Michel, and Akshay Venkatesh. The distribution of closed geodesics on the modular surface, and Duke’s theorem. *Enseign. Math. (2)*, 58(3–4):249–313, 2012. doi: 10.4171/LEM/58-3-2.
- [11] Ilya Khayutin. Arithmetic of double torus quotients and the distribution of periodic torus orbits. *Duke Math. J.*, 168(12):2365–2432, 2019. doi: 10.1215/00127094-2019-0016.
- [12] Yu. V. Linnik. *Ergodic Properties of Algebraic Fields*, volume 45 of *Ergebnisse der Mathematik und ihrer Grenzgebiete*. Springer-Verlag, Berlin–Heidelberg–New York, 1968. Translated from the Russian by M. S. Keane.
- [13] OpenAI. Equidistribution of Prime-Degree Torus Packets with Arbitrary Local Type. OpenAI Math Release preprint
[OAI:Equidistribution-of-Prime-Degree-Torus-Packets-with-Arbitrary-Local-Type-September-24-2026](#), 2026.
- [14] OpenAI. Equidistribution of primitive quartic torus packets for arbitrary orders. OpenAI Math Release preprint
[OAI:Equidistribution-of-Primitive-Quartic-Torus-Packets-for-Arbitrary-Orders-October-5-2026](#), 2026.
- [15] Paul Pollack. Nonnegative multiplicative functions on sifted sets, and the square roots of -1 modulo shifted primes. *Glasg. Math. J.*, 62(1):187–199, 2020. doi: 10.1017/S0017089519000041. Corrected author version available at <https://www.pollack-math.net/brunmultiplicative4.pdf>.
- [16] V. A. Rokhlin. Lectures on the entropy theory of measure-preserving transformations. *Russian Math. Surveys*, 22(5):1–52, 1967. doi: 10.1070/RM1967v022n05ABEH001224.
- [17] Peter Shiu. A Brun–Titchmarsh theorem for multiplicative functions. *J. Reine Angew. Math.*, 313:161–170, 1980. doi: 10.1515/crll.1980.313.161.
- [18] Harold M. Stark. Some effective cases of the Brauer–Siegel theorem. *Invent. Math.*, 23(2):135–152, 1974. doi: 10.1007/BF01405166.
- [19] Andreas Wieser and Pengyu Yang. A uniform Linnik basic lemma and entropy bounds. *Comment.*

